

Engineering Process of Polymer for Thermoelectric Applications

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Abstract Thermoelectric (TE) materials, which are capable of direct conversion between heat and electricity, offer a promising solution for sustainable energy harvesting. Conjugated polymers have emerged as compelling candidates for flexible TE devices owing to their intrinsic flexibility, low cost, and low thermal conductivity. The performance of polymer-based organic thermoelectrics (OTEs) is profoundly influenced by the processing methods, which dictate molecular packing, crystallinity, and film morphology. This review systematically summarizes recent advances in polymer processing techniques for TE applications, including solution processing, patterning techniques, and large-area fabrication. We discuss the interrelationships among processing techniques, polymer microstructure, and TE performance, concluding with the current challenges and future perspectives for industrializing high-performance OTE devices.

Keywords Organic semiconductors; Processing techniques; Organic thermoelectric devices; Thermoelectric performance

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INTRODUCTION

Conventional energy sources face increasingly severe sustainability challenges owing to their non-renewable nature, environmental pollution, and low energy conversion efficiency. In response to these issues, the global research community is actively developing clean alternative energy sources, such as solar, wind, and tidal power.^[1–4] Against this backdrop, thermoelectric (TE) materials have emerged as a significant research direction in the field of energy materials owing to their unique TE conversion properties. These materials enable direct conversion between heat and electricity based on the Seebeck effect and Peltier effect, offering advantages such as environmental friendliness and renewable resource utilization. Benefiting from continuous breakthroughs in material performance, derivative applications, including TE power generation devices, solid-state cooling systems, and self-powered sensing technologies, are advancing towards industrialization, providing innovative solutions for the green transformation of energy structures and the development of energy supply systems.

Organic thermoelectric (OTE) materials exhibit significant flexibility and tunability at the molecular level. Their diverse molecular sources and designability allow the systematic modulation of mechanical performance through molecular engineering (e.g., backbone optimization, side-chain design), blending, and micro/nano-structural control. Moreover, most

OTE materials possess good solubility, enabling high-throughput, large-area, and low-cost device fabrication via solution-based processes, such as spray coating, inkjet printing, and roll-to-roll techniques. These methods are simple, compatible with flexible substrates, and are suitable for complex curved surfaces and wearable applications. Furthermore, such materials demonstrate efficient TE conversion near room temperature, combined with intrinsic flexibility and low thermal conductivity (κ), making them particularly suitable for harvesting energy from body heat and environmental waste heat. They offer promising potential as sustainable self-powered solutions for next-generation wearable electronics.^[5,6] Recent advances in OTE highlight significant progress in both performance and application adaptability. In terms of performance, Tang *et al.* synthesized the n-type polymer poly(benzodifurandione) (PBFDO), demonstrating remarkable stability and high electrical conductivity (σ).^[7] Wang *et al.* developed a polymeric multi-heterojunction (PMHJ) to achieve a figure of merit (ZT) as high as 1.28.^[8] On the application front, the integration of high TE performance with mechanical flexibility is a key pursuit. Zhao *et al.* improved the stretchability of PBFDO films by incorporating poly(ethylene glycol) (PEG) as a plasticizer, which facilitated stress dissipation through chain slippage.^[9] This approach enabled the film to maintain its functionality under 100% strain, while σ increased from about 900 S·cm^{−1} to 2265 S·cm^{−1}. In parallel development, Liu *et al.* proposed one of the first n-type TE elastomers by integrating uniform bulk nanophase separation, thermally activated crosslinking, and targeted doping to develop a TE rubber that simultaneously achieves a high ZT, high stretchability with elastic recovery, and low modulus.^[10] This breakthrough ad-

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dresses the challenge of balancing the mechanical, electrical, and thermal properties of TE materials.

Although polymeric materials show great promise for flexible TE devices, the device fabrication processes remain challenging. The regulation of molecular packing, film morphology, and device structures requires various processing techniques. The processing strategy plays a critical role in determining the molecular orientation, crystallinity, and morphology, which directly influence the TE performance.^[11–15] Accordingly, this review provides representative processing methods developed in recent years (Fig. 1).^[8,16–26] Given the pivotal role of processing techniques in bridging molecular design and macroscopic performance, this review first distills the strategic processing paradigms essential for modulating polymer TE properties. Building upon this foundational framework, we systematically evaluate various processing technologies and provide an in-depth analysis of how they influence device performance by precisely controlling the microstructure.

CRITICAL ROLE OF PROCESSING TECHNIQUES IN THERMOELECTRIC PERFORMANCE

To evaluate the performance of TE materials, ZT is commonly used as a metric, which is determined by the σ , Seebeck coefficient (S), κ , and absolute temperature (T).

$$ZT = \frac{\sigma \cdot S^2}{\kappa} \cdot T \quad (1)$$

Based on these fundamental descriptions, it is clear that an excellent TE material should possess high σ and S but low κ . Although the intrinsic chemical structure of OTE polymers lays the foundation for their performance, the final device performance is largely determined by the micro- and macrostructures induced during processing. Processing techniques serve as the "critical process bridge" connecting molecular design with macroscopic device functionality. They dictate the transport behavior of charge carriers and phonons

through precise control over the polymer chain packing, orientation, crystallinity, and film morphology. Therefore, understanding and optimizing processing strategies are crucial for realizing high-performance practical OTE materials and devices.

The selection and optimization of processing methods ultimately serve the fundamental purpose of achieving decoupling and synergistic enhancement of TE parameters. Processing not only influences individual parameters but also employs strategies such as controlling molecular orientation and designing heterointerfaces to break the traditional coupling between σ and S or to reduce κ while maintaining electrical performance. Such proactive design and engineering of the microstructure serve as the driving force for continuously advancing performance benchmarks in OTE. Processing technologies exhibit a multilevel and integrated development trend, progressing "from microstructural regulation to macroscopic device manufacturing". At the micro level, basic solution-processing techniques (e.g., drop casting and spin coating) serve as the starting point for exploring the intrinsic properties of materials and structure-performance relationships. To achieve more precise functionalities, micro/nanopatterning techniques (e.g., photo-induced doping, nanoimprinting, and electron-beam lithography) have been developed to define high-resolution doping regions or structures within thin films, which are essential for improving device density and performance. Finally, for practical applications, large-area fabrication techniques (e.g., printing, spray coating, and roll-to-roll processing) and fiber/textile processing technologies are dedicated to addressing the challenges of scalability, flexibility, and wearable integration, representing the necessary path towards commercialization.

ADVANCED PROCESSING TECHNIQUES FOR POLYMER THERMOELECTRIC

Processing techniques serve as a critical link between the intrinsic

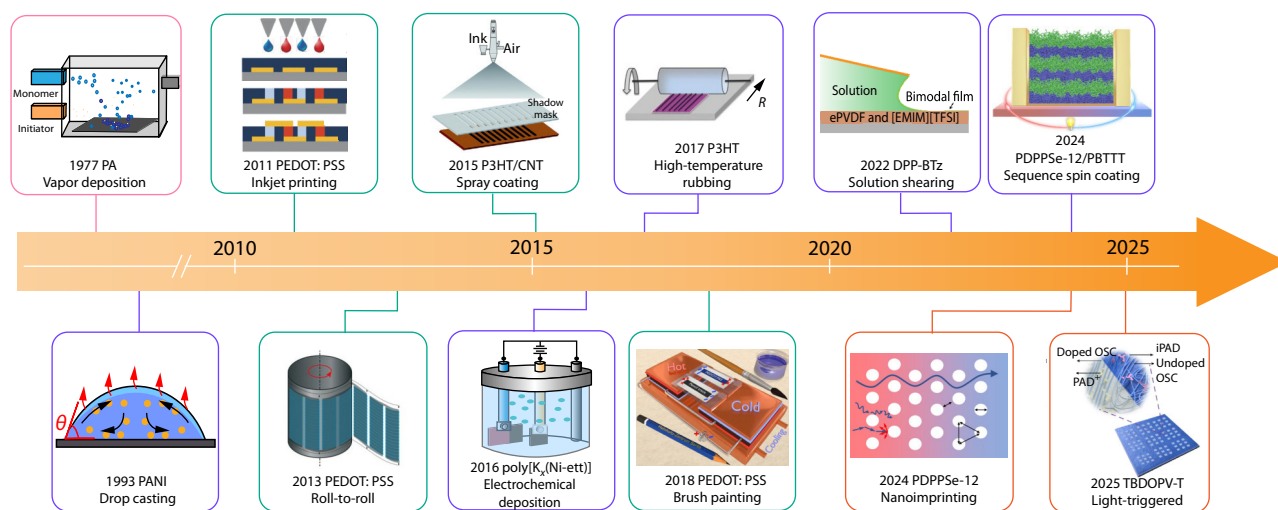


Fig. 1 Representative processing techniques for organic thermoelectric materials. (Reproduced with permission from Ref. [18]; Copyright (2011), Springer Nature; Reproduced with permission from Ref. [19]; Copyright (2013), Open Access; Reproduced with permission from Ref. [20]; Copyright (2015), The Royal Society of Chemistry; Reproduced with permission from Ref. [22]; Copyright (2017), Wiley; Reproduced with permission from Ref. [23]; Copyright (2018), Open Access; Reproduced with permission from Ref. [24]; Copyright (2022), Wiley; Reproduced with permission from Ref. [8]; Copyright (2024), Springer Nature. Reproduced with permission from Ref. [26]; Copyright (2025), Springer Nature.)

sic properties of materials and device performance. Based on their implementation approaches and application dimensions, processing techniques can be categorized into three major classes: solution-phase thin-film deposition techniques, micro/nano-patterning techniques, and large-area fabrication techniques. We systematically elaborate on the principles, advantages, limitations, and mechanisms influencing TE performance for each technique.

Solution-phase Deposition Techniques

The fabrication of high-performance OTE films from conjugated polymers (Fig. 2) fundamentally depends on the precise control of the transformation of the material from a dissolved, solvated state in a liquid medium to a solid-state thin film with a targeted microstructure. This critical "solution-to-solid" transition dictates the final molecular packing, crystallinity, domain size, and overall morphology, which are decisive factors for charge and phonon transport. The outcome of this process is governed by the interplay of solution physicochemical parameters and dynamic events during deposition and solvent removal. Key factors include solvent properties such as polarity, boiling point, volatility, capacity to solvate or induce pre-aggregation of polymer chains, which directly influence crystallization kinetics and molecular orientation. Concurrently, fluid dynamic phenomena during deposition, including capillary flows, Marangoni convection, and evaporation-induced self-assembly, can either lead to undesired nonuniformities or be harnessed to promote long-range order. Therefore, the rational manipulation of these parameters through solvent engineering, control of evaporation kinetics, and application of external fields constitutes the foundational strategy for tailoring film microstructure and optimiz-

ing TE performance.^[27]

Drop casting is one of the most fundamental and widely adopted solution processing techniques because of its operational simplicity and low cost (Fig. 3a). However, achieving control over the final film microstructure and uniformity remains inherently challenging, as both are dictated by complex and often non-uniform solvent evaporation dynamics. At the microscopic scale, the competition between solvent evaporation rate and self-assembly kinetics of the solute governs the degree of molecular order and crystallinity. On a macroscopic level, differential evaporation rates across the droplet are typically faster at edge-drive capillary flows that lead to non-uniform material deposition, most prominently observed as the "coffee-ring effect."

Therefore, the deliberate regulation of evaporation kinetics is central to addressing the challenges of drop casting, encompassing strategies from solvent selection to environmental control. The choice of the solvent itself is a primary lever, directly coupling the evaporation behavior with molecular self-assembly. For instance, Huq *et al.* demonstrated that solvents with high solubility and low volatility promoted a predominantly edge-on orientation in poly(3-hexylthiophene) (P3HT) films. Beyond intrinsic solvent properties, the drying environment can be actively engineered.^[28] Chen *et al.* implemented a solvent vapor-enhanced drop-casting (SVE-DC) technique, which creates a vapor-rich atmosphere to slow down and homogenize the evaporation process.^[29] These controlled drying kinetics guided the assembly of 3,6-bis-(thiophen-2-yl)-*N,N'*-bis(2-octyl-1-dodecyl)-1,4-dioxo-pyrrolo[3,4-*c*]pyrrole and thieno[3,2-*b*]thiophene (PDBT-TT)

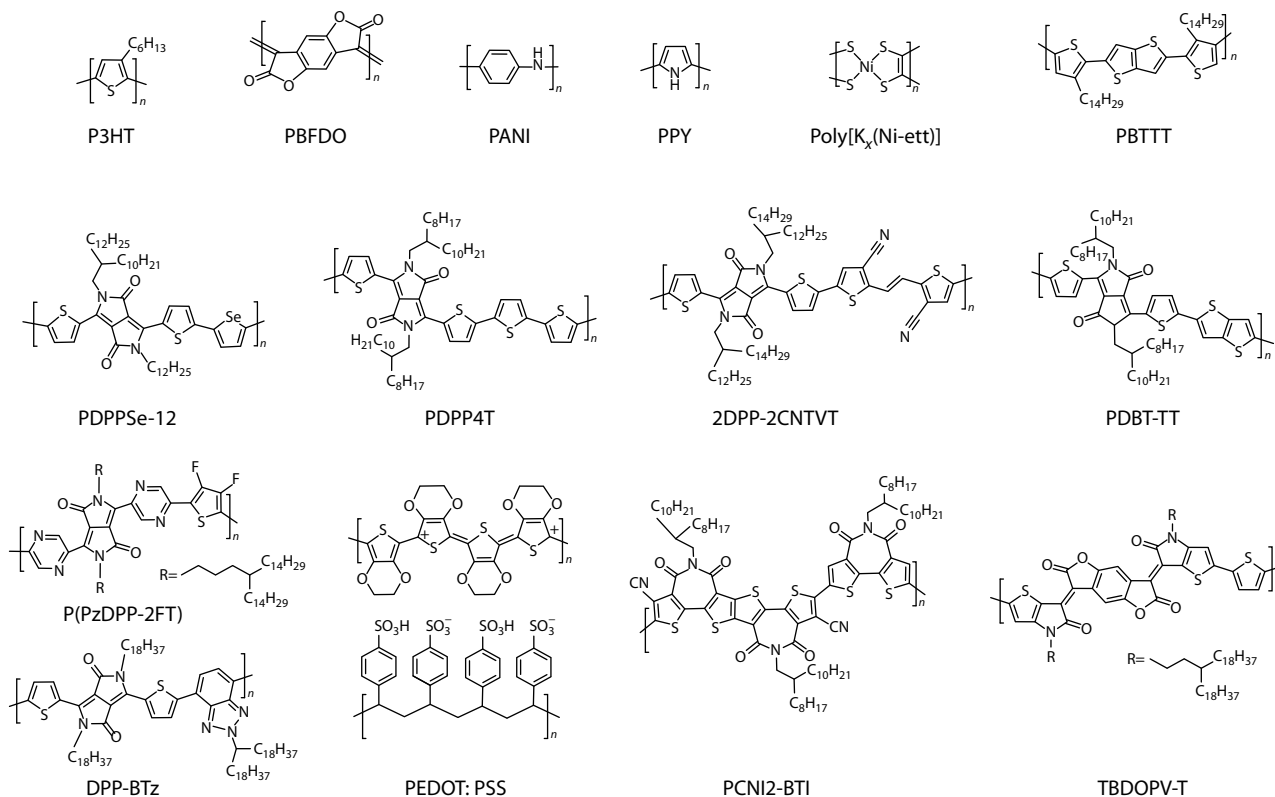


Fig. 2 Molecular structures of conjugated polymers in this review.

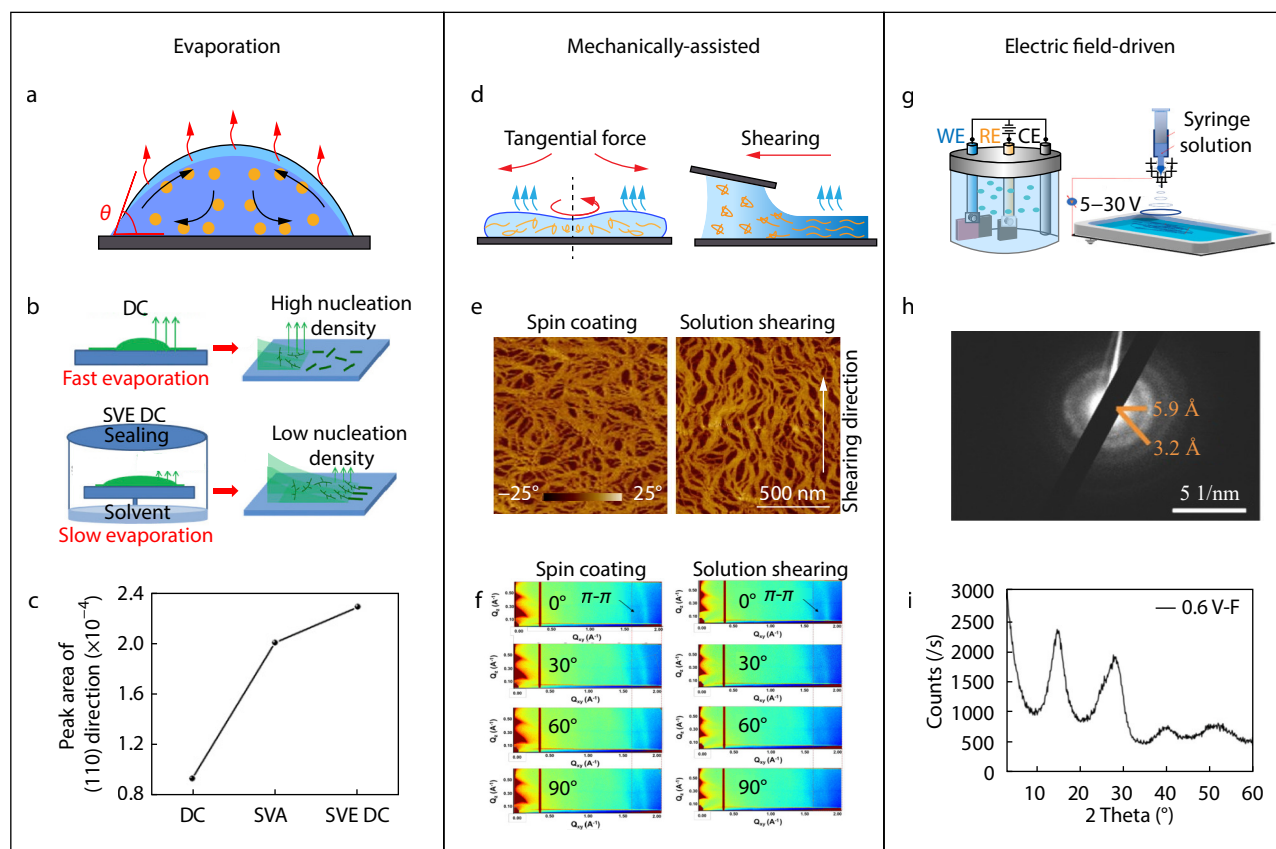


Fig. 3 Schematic diagrams of solution-phase film deposition techniques. (a) Schematic of drop-casting (DC) process driven by solvent evaporation; (b) Schematic comparing standard drop casting and solvent vapor-enhanced drop casting (SVE-DC); (c) Comparison of crystallinity peak area in the (100) direction for films prepared by different methods; (b, c: reproduced with permission from Ref. [29]; Copyright (2016), Wiley.) (d) Schematic of mechanically-assisted solution processing; (e) AFM phase images of films prepared by spin coating and solution shearing; (f) GIWAXS patterns of spin-coated and solution-sheared films; (e, f: reproduced with permission from Ref. [35]; Copyright (2019), Springer Nature.) (g) Schematic of electric-field-driven solution processing (reproduced with permission from Ref. [43]; Copyright (2024), Elsevier.); (h) Selected-area electron diffraction pattern of poly[K_x(Ni-ett)] film. The ring pattern indicates the polycrystalline characteristics of the film; (i) Powder X-ray diffraction patterns of poly[K_x(Ni-ett)] films after being scraped off and ground into powder. (h, i: reproduced with permission from Ref. [21]; Copyright (2016), Wiley.)

molecules into a highly aligned, crystalline film with fibrous order (Fig. 3b), resulting in a quantitatively larger crystallinity peak area in the (100) direction compared with films from standard drop casting (Fig. 3c).

Furthermore, coating techniques can be utilized to fabricate TE composite materials such as carbon nanotubes/polyaniline (CNTs/PANI) composite films. Li *et al.* demonstrated that modulating the doping level could optimize the microstructure and enhance the TE properties of CNTs/PANI composites.^[30] Further investigations revealed that sequential dedoping-redoping treatments effectively regulated the interfacial structure of CNTs/PANI composite films, leading to significant improvements in their TE performance.^[31]

Evaporation-driven processing approaches have evolved to incorporate mechanical assistance into solution processing techniques. Spin coating and solution shearing are representative methods in this category (Fig. 3d). Spin coating utilizes radially symmetric tangential forces generated by substrate rotation to spread the solution and accelerate solvent evaporation, thus favoring the formation of films with uniform thickness and a microscopically isotropic structure. Yoo *et al.*

studied the effect of the rotation speed on *PF* and found that as the revolutions per minute increased from 300 to 1500, the thickness of the film decreased from 357 nm to 89 nm and the roughness decreased from 2.097 nm to 0.726 nm.^[32] However, excessive rpm may induce phase separation in poly(3,4-ethylenedioxythiophene) polystyrene sulfonate (PEDOT:PSS) film, resulting in a decrease in *PF*. Zhang *et al.* developed a modified spin coating method utilizing on-the-fly dispensing to produce ultrathin films of poly[2,5-bis(3-dodecylthiophen-2-yl)thieno[3,2-*b*]thiophene] (PBTTC-12) with a precise thickness of 1.6 nm.^[33]

In contrast, solution shearing applies unidirectional shear forces *via* a blade or slot-die to directly induce ordered alignment of polymer backbones along the shearing direction during solution spreading and drying, thus producing thin films with well-defined molecular orientation. This orientation directly governs the charge-transport channels, electronic structures, and intermolecular interactions in polymer semiconductors, thereby potentially influencing the associated TE parameters (Fig. 3d). Feng *et al.* reported that the σ of the cyano-functionalized imide-based fused bithiophene imide

dimer (PCN12-BT1) film prepared by spin coating was $89.2 \text{ S}\cdot\text{cm}^{-1}$, while the σ of the film by solution shearing reached $150.2 \text{ S}\cdot\text{cm}^{-1}$.^[34] Xu *et al.* used a blade with line structures to induce ordered chain alignment along the shear direction. By adjusting the evaporation rate of the solvent, solution concentration, shear rate, and temperature, they produced a solution-sheared film with an ordered chain alignment that exhibited a three-fold enhancement in carrier mobility (μ) compared to spin-coated films.^[35] Fig. 3(e) shows that the nanofibers in the solution-shearing film are more orderly aligned compared to those in the spin-coated film. Fig. 3(f) also shows that the solution-shearing film has a significantly more intense edge-on π - π stacking peak along the shearing direction than in the perpendicular direction, but this phenomenon is not observed in the spin-coating film. In solution shearing, uneven solvent evaporation at the meniscus edge may also drive a capillary flow that carries solutes outward, resulting in nonuniform film deposition and impaired charge transport. Yang *et al.* introduced a guest solvent n-hexane, which has a higher boiling point and lower surface tension, into the host solvent chloroform (CF) to promote crystal growth during film formation.^[36] This transformation resulted from faster P3HT crystallization coupled with slower solvent evaporation, leading to a more compact molecular assembly with shorter π - π stacking distances. The σ and the PF of P3HT-(mixed solvent) increased to $605.39 \text{ S}\cdot\text{cm}^{-1}$ and $204.62 \mu\text{W}\cdot\text{m}^{-1}\cdot\text{K}^{-2}$, respectively, while the σ and the PF of P3HT-(CF) were only $141.38 \text{ S}\cdot\text{cm}^{-1}$ and $47.98 \mu\text{W}\cdot\text{m}^{-1}\cdot\text{K}^{-2}$, respectively.

Furthermore, the electric-field-driven approach enables precise control over material growth, *in situ* doping, and direct patterning capabilities. Electrochemical deposition typically employs a three-electrode system in which a specific potential or current is applied to the working electrode to drive the oxidation or reduction reactions of monomers or precursors in the solution (Fig. 3g).^[37–41] By adjusting parameters such as potential, current density, and electrolyte composition, this method finely regulates the composition of the material, doping concentration, crystallinity, and microstructure, thereby directly influencing its TE properties. Sun *et al.* transformed amorphous poly[$\text{K}_x(\text{Ni-ethylenetetra-thiolate})$] (poly[$\text{K}_x(\text{Ni-ett})$]) powder into a polycrystalline thin film *via* electrochemical deposition.^[21] Fig. 3(h) shows the rings of selected-area electron diffraction, indicating a polycrystalline film. The powder X-ray diffraction (XRD) of poly[$\text{K}_x(\text{Ni-ett})$] showed broad peaks at 14.8° and 27.8° , which also matched the selected-area electron diffraction in Fig. 3(i). The polycrystalline film exhibited a remarkable improvement in σ , ultimately achieving an outstanding PF of $453 \mu\text{W}^{-1}\cdot\text{m}^{-1}\cdot\text{K}^{-2}$. It also offers a pathway for simultaneous film fabrication and patterning through mask-assisted processes. For instance, Liu *et al.* successfully deposited a patterned poly[$\text{K}_x(\text{Ni-ett})$] film using a polydimethylsiloxane (PDMS) mask to fabricate an array of 6×18 rectangular bars in a single step.^[42] The resulting device achieved a maximum power density of $577.8 \mu\text{W}\cdot\text{cm}^{-2}$ under a temperature difference of 12 K. Electric field can also be employed to induce the formation of nanometer-scale fibers from a polymer solution, thereby enabling the fabrication of nanofibers with favorable mechanical properties and a

porous structure. Codau *et al.* fabricated TE nanocomposites based on PEDOT: PSS, multi-wall carbon nanotubes, and thermoplastic polyurethane *via* wet-electrospinning, which exhibited PF of $4.28 \mu\text{W}\cdot\text{m}^{-1}\cdot\text{K}^{-2}$, σ of $85 \text{ S}\cdot\text{m}^{-1}$, and a strain range of up to 190%.^[43]

Micro/Nano-patterning Techniques

Patterning focuses on precisely regulating the localized properties of materials at the micro- and nanoscale, representing an essential pathway towards high-performance and functionally integrated devices.^[44,45] Depending on the working principle, the main approaches include photo-induced patterning, nanoimprinting, electron-beam lithography, and emerging methods, such as nanoconfined electrochemical ion implantation.^[46,47]

Photo-induced patterning is a technique that utilizes irradiation with light of specific wavelengths to selectively excite materials or dopant molecules using photon energy. This process induces phenomena such as electron transfer, chemical bond reorganization, or phase transitions, triggering localized photochemical reactions (*e.g.*, photo-doping, photocrosslinking, or photo-decomposition) in the material. Consequently, it enables precise modulation of the microstructure and properties within micro-regions, ultimately achieving targeted and controllable tuning of the electrical performance in specific areas. This approach overcomes the limitations imposed by intrinsic energy-level mismatches in traditional chemical doping, allowing for efficient doping even with weakly active dopants.^[48] Wang *et al.* designed a series of inactive photoactive dopants, which achieved photoinduced patterned doping of various n-type organic semiconductors, reaching a σ of up to $30 \text{ S}\cdot\text{cm}^{-1}$ and a feature resolution of $1 \mu\text{m}$.^[26] Ji *et al.* patterned poly[2,5-bis(2-octyldodecyl)pyrrolo[3,4-*c*]pyrrole-1,4(2*H*,5*H*)-dione-3,6-diyl)-*alt*-(2,2', 2'': 5', 2'': 5'', 2'': quaterthio-phen-5,5''-diyl)] (PDPP4T) using a photoexcitation-assisted molecular doping method (Fig. 4a).^[49] This approach overcomes the limitations imposed by energy-level misalignment by facilitating efficient photoinduced electron transfer between the polymer and dopants (Fig. 4b). Optical microscope images of various doped patterns clearly distinguished between the doped and undoped regions of the lines, demonstrating a patterning resolution of $1 \mu\text{m}$ with lateral scalability (Fig. 4c). Xue *et al.* utilized photo crosslinker to achieve patterning of PDPP4T material, achieving sub-500 nm resolution while maintaining a high σ of $30 \text{ S}\cdot\text{cm}^{-1}$.^[50]

Additionally, high-precision patterning of polymer films can be achieved using nanoimprint lithography (Fig. 4d). Nanoimprint lithography enables direct fabrication of nanoscale patterns on polymer films through a physical imprinting mechanism. This technique is not constrained by the diffraction limit imposed by the light wavelength in traditional optical lithography, thereby allowing for a higher precision in pattern transfer. Patterned films produced *via* nanoimprint lithography can effectively modulate the κ of the materials. For example, Zhao *et al.* utilized this method to fabricate periodically porous polymer films.^[25] Under fixed periodic conditions, increasing the pore size enhanced the phonon-like scattering effects, further suppressing κ (Fig. 4e). Compared to the non-porous selenium-substituted diketopyrrolopyrrole (PDPPSe-12) films, the porous PDPPSe-12 films exhibited

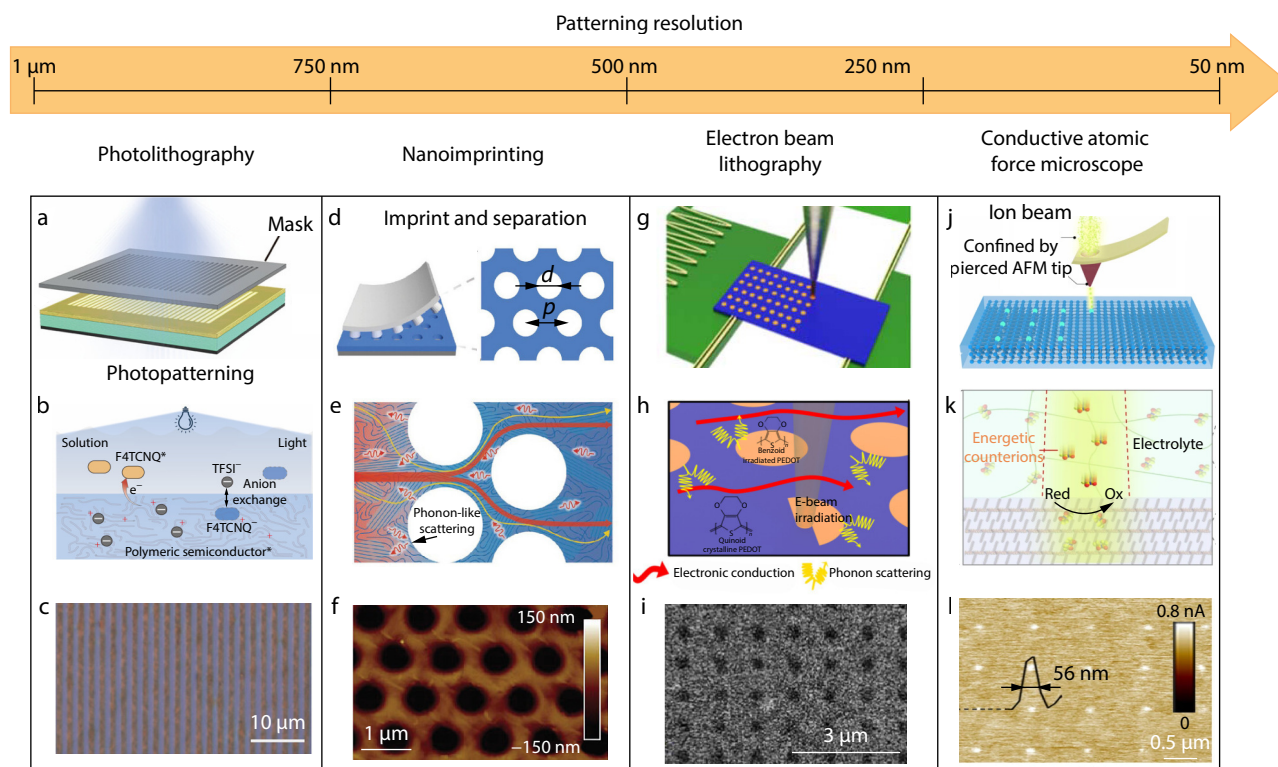


Fig. 4 Schematic representations of patterning techniques. (a) The photopatterned doping process; (b) Schematic of photopatterning in doped PDPP4T through photoexcitation assisted molecular doping; (c) Optical microscope image of lines with a width of 1 μm ; (a–c: reproduced with permission from Ref. [49]; Copyright (2024), Open Access.) (d) Schematic diagram of porous film fabrication process using nanoimprinting method; (e) Schematic illustration of phonon transport in a porous structure; (f) AFM images of the porous films; (d–f: reproduced with permission from Ref. [25]; Copyright (2024), Wiley.) (g) Schematic of fabrication process of nanopatterned H-PEDOT film via electron beam lithography; (h) Schematic of the electron and phonon transport in the film; (i) An SEM image of an H-PEDOT sample with a nanopatterned sample; (g–i: reproduced with permission from Ref. [52]; Copyright (2024), Wiley.) (j) Schematic diagram of single-ion implantation for inorganic electronics; (k) Schematic diagram showing high-spatial resolution nanoconfined electrochemical ion implantation (NEII) doping; (l) Conductive AFM image of doped PBTTT films with 56 nm doping resolution. (j–l: reproduced with permission from Ref. [53]; Copyright (2024), Springer Nature.)

a significant reduction in κ while maintaining efficient charge transport and a nearly unchanged PF . When the period was set to 1 μm , the pore diameter to 652 nm, and the porosity to 25.5% (Fig. 4f), the κ of the film decreased by 57%, and the ZT increased to 0.47 ± 0.05 , which is 1.7 times higher than that of the non-porous film.

Electron beam lithography (EBL) is a direct-write micro/nanofabrication method that does not require traditional photomasks or photoresists (Fig. 4g). It uses a focused high-energy electron beam to perform scanning exposure on the material surface. Interactions between the electrons and the material, such as chemical bond breaking, cross-linking, local thermal effects, or structural phase transitions, directly alter the physical or chemical properties of the exposed areas, thereby enabling patterning.^[51] Lee *et al.* used electron beams (without any corrosion-resistant layers) to directly write patterns with a resolution of several tens of nanometers on H-PEDOT thin films.^[52] Under e-beam irradiation, PEDOT:PSS films underwent a structural transition from quinoid to benzoid. The incorporation of benzoid units into the optimized patterned configuration significantly reduced κ (Fig. 4h). By adjusting the space between the dots in the pattern array (Fig. 4i), the κ of the film was reduced from 8.51

$\text{W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ to $2.48 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ without a significant decrease in σ .

In addition to these categories, several additional patterning methods have shown promise for OTE applications. Xiang *et al.* presented a method for patterning and doping based on nanoconfined electrochemical ion implantation (NEII) (Figs. 4j and 4k).^[53] In this method, a conductive atomic force microscopy tip served as the counter electrode, enabling the patterning of various polymers (e.g., poly(2,5-bis(3-tetradecylthiophen-2-yl)thieno[3,2-*b*]thiophene) (PBTTT) and P3HT) via ion implantation. This method achieved a maximum resolution of PBTTT film reach 56 nm (Fig. 4l). This technology further achieves nano-scale patterning resolution and provides research pathways for constructing high-performance nano-devices.

Large-area Fabrication Technologies

Large-area fabrication technology is essential for the industrialization of OTE materials, with the core objective being the manufacturing of devices that exhibit uniform performance and are suitable for scalable production.^[54] Common methods include spray coating, inkjet printing, screen printing, brush painting, and roll-to-roll processing. Furthermore, fiber and textile technologies are emerging as promising routes for producing weav-

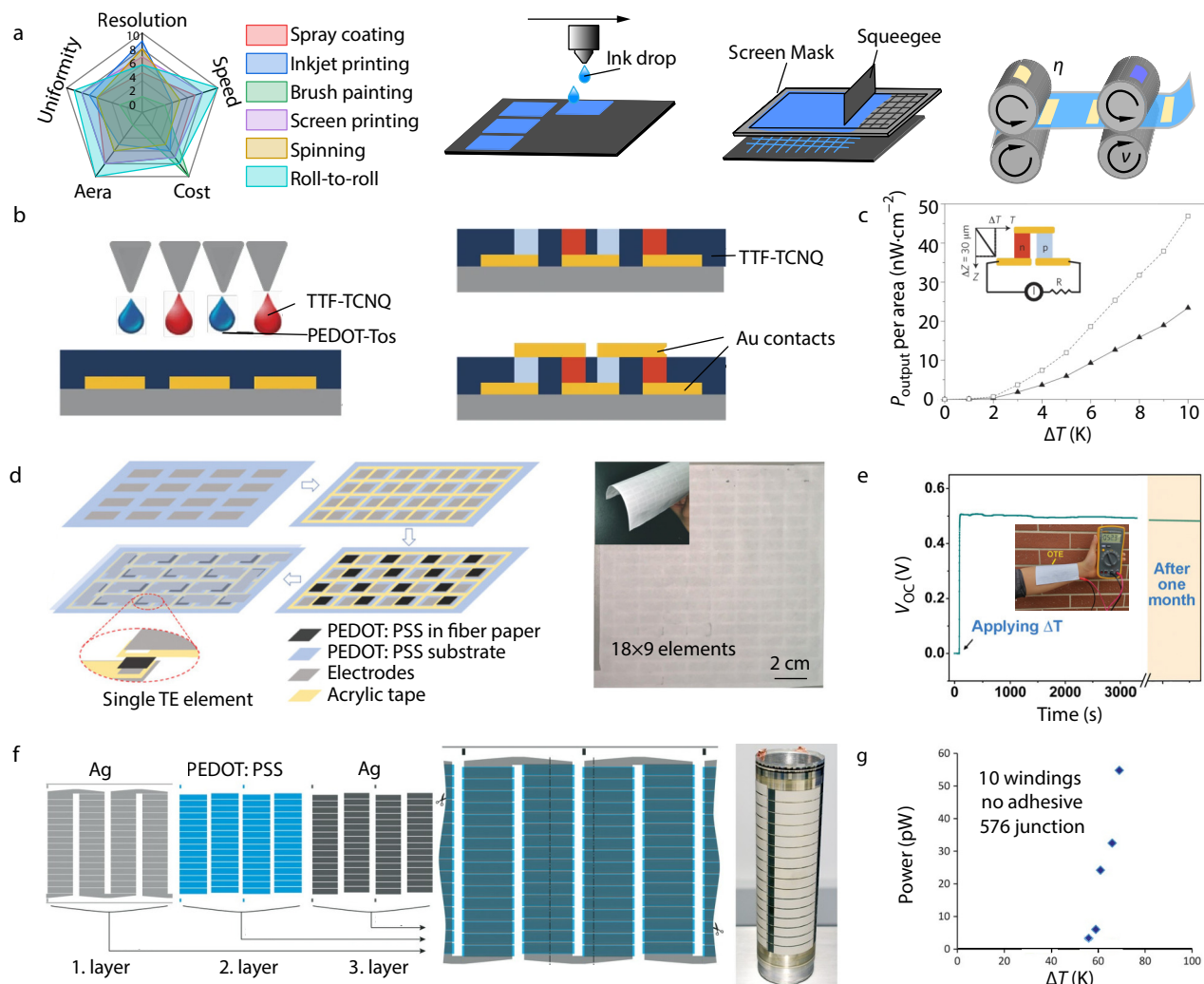


Fig. 5 Schematic diagrams of large-area fabrication techniques. (a) Radar charts for comparison of different large-area fabrication techniques, and schematics of inkjet printing, screen printing, and roll-to-roll processing; (b) Schematic diagrams of TE device fabrication via inkjet printing; (c) Maximum power output per area of the TEGs consisting of 54 thermocouples; (b, c: reproduced with permission from Ref. [18]; Copyright (2011), Springer Nature.) (d) Schematic illustration of the screen printing process of an OTE generator with PEDOT:PSS legs and a photograph of the integrated generator with 162 legs; (e) The durability test of an OTE array under a temperature difference of 85.5 K; (d, e: reproduced with permission from Ref. [63]; Copyright (2019), Wiley.) (f) Illustration of the roll-to-roll process showing the pattern of each printed layer and the final device architecture and illustration of and picture of the rolled-up final device; (g) The output power of the device. (f, g: reproduced with permission from Ref. [19]; Copyright (2013), Open Access.)

able and knittable TE modules. Fig. 5(a) compares the performance of these methods in terms of resolution, uniformity, applicable area, cost, and processing speed using a radar chart.

Spray coating utilizes compressed air, electric fields, or ultrasonic waves to atomize a solution into fine droplets, which are then deposited on a substrate. The complete process typically involves stages, such as solution atomization, droplet transport, deposition, spreading, and drying. Key parameters, including solution concentration, nozzle traverse speed, nozzle-to-substrate distance, and substrate temperature, significantly influence the surface uniformity and performance of the films. Spray coating is particularly suitable for the rapid fabrication of large-area coatings with consistent thickness.^[55] For example, Lin *et al.* adjusted a series of spraying conditions and fabricated an OTE device with an output power of 12.1 nW·cm⁻² with a spraying distance of 14 cm,

substrate temperature of 80 °C, spraying time of 2 s, and spraying interval of 30 s.^[56] This technique is particularly suitable for integration with structurally complex substrates, offering strong technical support for the industrial production of flexible electronics. Darabi *et al.* sprayed poly(benzimidazobenzanthroline) and poly(ethylene imine) onto a single fiber to fabricate n-type conductive fibers with σ of $(8 \pm 2) \times 10^{-3}$ S·cm⁻¹ and S of $-79 \mu\text{V}\cdot\text{K}^{-1}$.^[57] Furthermore, Sun *et al.* employed mask-assisted spray doping to fabricate functional fibers with p-n junction structures, which were subsequently woven into textiles to achieve a power density of 70 mW·m⁻².^[58] Chang *et al.* proposed a dual-regime spray system method to achieve immediate *in situ* polymerization of polypyrrole.^[59] This approach utilizes two distinct gas flows: a low-speed flow to mix the oxidant and pyridine solution and a high-speed flow for focused spraying. These studies high-

light the significant value of spray coating in the large-scale preparation of composite materials, complex patterning, and fabrication of wearable TE devices.

Inkjet printing is a digitally controlled, non-contact material deposition technology that achieves patterning through precise control of the selective release and deposition of microdroplets at specific locations. Compared to spray coating, inkjet printing offers distinct advantages, such as high resolution, site-specific controllability, and high material utilization, making it particularly suitable for fabricating complex patterns and fine structures. However, this technique also imposes stringent requirements on ink properties (e.g., viscosity and surface tension) and may present challenges in achieving a uniform film thickness.^[60] Bubnova *et al.* fabricated a thermoelectric generator (TEG) with 54 legs by using inkjet printing.^[18] The device exhibited a maximum power output of 0.128 μW at a temperature difference of 10 $^{\circ}\text{C}$, with a potential power density of 0.27 $\mu\text{W}\cdot\text{cm}^{-2}$ at a 30 $^{\circ}\text{C}$ temperature difference (Figs. 5b and 5c). Parameters such as ink concentration, viscosity, and particle size are crucial for the print quality. Ferhat *et al.* summarized their approach to regulating PEDOT-based ink in three key aspects: ensuring fluid jettablity to prevent nozzle clogging, optimizing the droplet formation process for improved precision and quality, and modulating substrate adhesion to suppress ink spreading.^[61] By adjusting the ink viscosity and surface tension to 5 $\text{mPa}\cdot\text{s}$ and 31 $\text{mN}\cdot\text{m}^{-1}$, respectively, they successfully printed the target pattern and achieved a power density of 0.266 $\mu\text{W}\cdot\text{cm}^{-2}$ at a ΔT of 20 K. In addition to the ink properties, the choice of substrate also influences the printing quality and process stability of inkjet printing. Jing *et al.* demonstrated that substrates with low roughness and good wettability, such as silicon oxide, yield films with higher σ ($300\pm 10\text{ S}\cdot\text{cm}^{-1}$) compared to poorly wetting substrates like parylene ($265\pm 10\text{ S}\cdot\text{cm}^{-1}$).^[62]

Unlike the non-contact nature of spray coating and inkjet printing, screen printing employs a mesh stencil to force viscous ink through predefined patterns, thereby transferring the design onto a substrate. Owing to its resolution limitations, this method is more suitable for applications where high pattern precision is not critical but rapid and large-area coverage is required, such as substrate coating for flexible wearable TE devices or simple module integration. Zheng *et al.* fabricated a paper-based TE device comprising 162 legs using screen printing and achieved a maximum output power of 0.32 μW .^[63] Notably, the device maintained an open-circuit voltage of 0.5 V after one month of operation (Figs. 5d and 5e). Similarly, Wei *et al.* fabricated a device with an output power of 50 μW at ΔT of approximately 100 K by printing PEDOT: PSS onto paper.^[64] The material versatility of screen printing enables fabrication of various composites, including PEDOT: PSS/Ag nanowire and PANI/Cu₃Se₂ films.^[65,66]

Brush painting involves manually applying coatings to substrates using brushes, with advantages including simple operation, rapid processing, low cost, and compatibility with non-planar substrates. Zhang *et al.* extended this method to different substrates (art paper, polyethylene terephthalate (PET), and textiles) using PDPPSe-12, achieving 14.7 nW maximum output power of paper.^[12] The thickness of the film can be controlled from 100 nm to micrometers by adjusting the solu-

tion concentration and number of coating repetitions. Brus *et al.* demonstrated this technique to fabricate PEDOT: PSS films on paper, achieving low-cost TEGs that generate over 10 nW at a ΔT of 50 K.^[23] Notably, Zhang *et al.* revealed that shear stress during brushing promotes polymer alignment.^[67] By optimizing the substrate temperature and brush speed ($0.5\text{ mm}\cdot\text{s}^{-1}$), they achieved a remarkable PF of $170\text{ }\mu\text{W}\cdot\text{m}^{-1}\cdot\text{K}^{-2}$ along the brushing direction.

Roll-to-roll technology provides a critical pathway for the industrialization of high-performance, low-cost, and large-area flexible TE devices. This technology enables large-area uniform film formation of TE materials through continuous coating or printing processes, which helps maintain spatial consistency in the material conductivity. Furthermore, its integrated production line can complete multiple steps, such as electrode printing, TE layer fabrication, and encapsulation, thereby significantly enhancing device manufacturing efficiency. Zhang *et al.* loaded p-type ink, n-type ink, and PEDOT: PSS aqueous solutions onto three separate rollers.^[68] Using continuous three-roller printing, they successfully fabricated complete OTE devices. Demonstrating the scalability of roll-to-roll, Søndergaard *et al.* fabricated 18000 serially connected silver/PEDOT: PSS junctions on a polyethylene terephthalate substrate by roll-to-roll, and serial devices with 576 junctions can achieve a voltage over 160 μV and power over 50 pW (Figs. 5f and 5g).^[19] In another study, Zhang *et al.* divided the roll-to-roll process into four steps: electrode printing, lamination, TE ink injection, and cover electrode printing.^[69] After optimizing the experimental conditions, they printed a flexible TEG with a width of 152.4 cm and length of 914.4 cm using this roll-to-roll process. The device with 20 p-n junctions achieved a maximum output power of 6.5 μW at a ΔT of 25 K. However, owing to the relatively fast processing speed, the material crystallinity, doping distribution, and interface structure may not be as ideal as those achieved in laboratory-scale slow processes. Therefore, it is necessary to synergistically optimize the ink formulation, drying kinetics, and process parameters to maintain high TE performance while ensuring high production capacity.

Fiber spinning is a processing technology that uses spinning equipment to stretch and solidify polymer solutions or melts into continuous fibers, making it an ideal approach for creating flexible, large-area, wearable electronic textiles. This technique directly produces one-dimensional fiber structures and leverages shear and stretching forces during the process to actively induce high alignment and orientation of polymer chains along the fiber axis, thereby enhancing the μ and σ . Zhang *et al.* found that appropriately reducing the solution concentration combined with a flow-enhanced crystallization method could promote the ordered arrangement and crystallization of polymer chains.^[70] This technique utilizes flow shear and stretching to disaggregate and pre-align polymer aggregates, enabling the spinning of various conjugated polymers into fibers and achieving continuous fiber production. The mechanical performance of conductive fibers and fabrics is crucial, prompting increasing efforts to enhance their stretchability. Sun *et al.* used coaxial microfluidic devices to fabricate helical PEDOT: PSS core fibers with a polyurethane (PU) sheath, creating PU@PVA-PEDOT: PSS

fibers with stretchability of up to 500%.^[71] Li *et al.* significantly enhanced the TE performance by improving the wet-spinning process, such as optimizing the dispersibility of CNTs, precisely controlling the doping level of polymers, enhancing the polymer alignment, and implementing effective post-treatment techniques.^[72–74] These strategies not only improved the mechanical properties and stability of the fibers but also advanced the practical applications of OTE materials in wearable devices, self-powered sensors, and sustainable energy harvesting, demonstrating broad prospects for development.

Each of the processing methods discussed above exhibits distinct characteristics: solution processing is noted for its operational simplicity and cost effectiveness; patterning techniques are indispensable for achieving high resolution and integration density; and large-area fabrication is key to scalable production. Despite significant progress, the processing of OTE materials still faces several challenges: the stability of film performance requires further improvement, precise microstructural control at scale remains difficult, and ensuring batch-to-batch consistency demands sophisticated process control and monitoring. The commercial viability of OTE devices ultimately depends on the establishment of stable, high-performance batch production processes.

CORRELATING PROCESSING, MICROSTRUCTURE, AND THERMOELECTRIC PERFORMANCE

The influence of different processing methods on the polymer microstructure is the key factor driving performance variations. The structure-function relationship of OTE materials exhibits distinct multi-scale characteristics, encompassing hierarchical organizations ranging from molecular structures and nanoscale assemblies to microdomains and macro-scale architectures. This multilevel synergy effectively optimizes charge transport behavior and enhances phonon scattering, thereby improving the overall TE performance. The impact of structural features on charge-transport mechanisms varies across micro-, meso-, and macro scales. At the molecular/nanoscale, molecular size, rigidity, and the introduction of solubilizing side chains influence local molecular packing, substrate interactions, and solubility, which in turn regulate the size of crystalline domains and phase separation behavior. By optimizing the molecular structure and chemical properties, precise control over the intermolecular interactions and local charge transfer can be achieved. At the mesoscale, spatial confinement and interface effects render grain size, orientation, and structural evolution as critical determinants of the carrier and phonon transport properties. Typically, μ within crystalline regions is significantly higher than that in the amorphous phases or grain boundaries. Although unfavorable molecular packing or large-scale grain boundaries restrict charge transport pathways, they concurrently enhance phonon scattering. At the macroscale, the size of structural crystalline domains and roughness of phase interfaces directly govern the transport efficiency of the carriers. This review examines how processing routes enhance σ , suppress κ , and decouple the relationship between σ and S from the perspective of microstructural modulation in polymer films.

Enhancing Electrical Conductivity

In terms of the molecular design of OTE materials, rigid and pla-

nar conjugated backbones are crucial for strengthening π - π interactions, improving crystallinity, and extending polaron delocalization lengths upon doping, all of which contribute to the enhanced σ and other TE performance. Molecular packing dictates the charge transport pathways, electronic structure, and intermolecular interactions within polymer semiconductors, directly influencing the key metrics of the TE performance. Researchers have recently developed a series of mechanical and templating approaches to regulate the aggregation structure in films, enhance the crystalline orientation and overall structural order, and thereby achieve superior σ and TE performance.

Feng *et al.* fabricated polymer films using spin coating and solution shearing, respectively (Fig. 6a).^[34] Their results demonstrated that solution-sheared films exhibit more ordered chain alignment, leading to higher μ and σ . Specifically, the solution-sheared films showed a μ of $0.44 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ and a σ of 150.2 S cm^{-1} , surpassing the values of the spin-coated films ($0.22 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ and 89.2 S cm^{-1} , respectively) (Fig. 6b). Beyond shear-induced alignment, other processing strategies can also be used to template molecular order. Liu *et al.* applied repeated freeze-thaw cycles to PEDOT:PSS solutions, where growing ice crystals acted as templates to promote ordered chain arrangement within the polymer-rich phase (Fig. 6c).^[75] After optimization of the processing rate and needle length, gel fibers subjected to two freeze-thaw cycles achieved a PF of $(34 \pm 2) \mu\text{W m}^{-1} \text{ K}^{-2}$ (Fig. 6d) and a σ of $(1013 \pm 32) \text{ S cm}^{-1}$, significantly higher than the initial value of about 200 S cm^{-1} . Increasing the number of film layers is also a method for enhancing the alignment ordering of polymer chains. Lee *et al.* alternately spin-coated PEDOT: PSS and PANI-CSA (camphorsulfonic acid) layers (Fig. 6e).^[76] They observed that the polymer chains in the upper layers became more stretched as the number of layers increased. The improved alignment of PEDOT:PSS and PANI-CSA chains combined with hole diffusion from PANI-CSA to the PEDOT:PSS layer enhanced the overall σ of the multilayer structure. As shown in Fig. 6 (f), the PF of a single-layer PEDOT:PSS film was $24 \mu\text{W m}^{-1} \text{ K}^{-2}$, while that of a five-layer ($n=5$) film reached $49 \mu\text{W m}^{-1} \text{ K}^{-2}$. Additionally, the multilayer film maintained its power generation capability at temperature up to 120°C .

Thermal Conductivity Engineering

The primary strategies for regulating the κ of conjugated polymers include modulating molecular packing modes to optimize intermolecular heat transfer pathways; controlling intra- and inter-chain interactions to alter the propagation characteristics of heat-carrying phonons; introducing scattering centers or nanostructures to enhance phonon scattering and reduce κ . Ideal polymeric materials with low κ typically possess the following characteristics: incorporation of heavy atoms such as oxygen, sulfur, and phosphorus into the backbone to reduce phonon group velocity, thereby decreasing κ ; utilization of asymmetric polar substituents and side-chain modifications to increase side-chain length, weaken intermolecular interactions, and enhance phonon scattering; and the introduction of scattering centers, including chain entanglements, packing disorder, structural defects, pores, and interfaces to effectively regulate thermal transport behavior without compromising critical properties such as charge transport.^[77,78]

The incorporation of porous structures creates additional

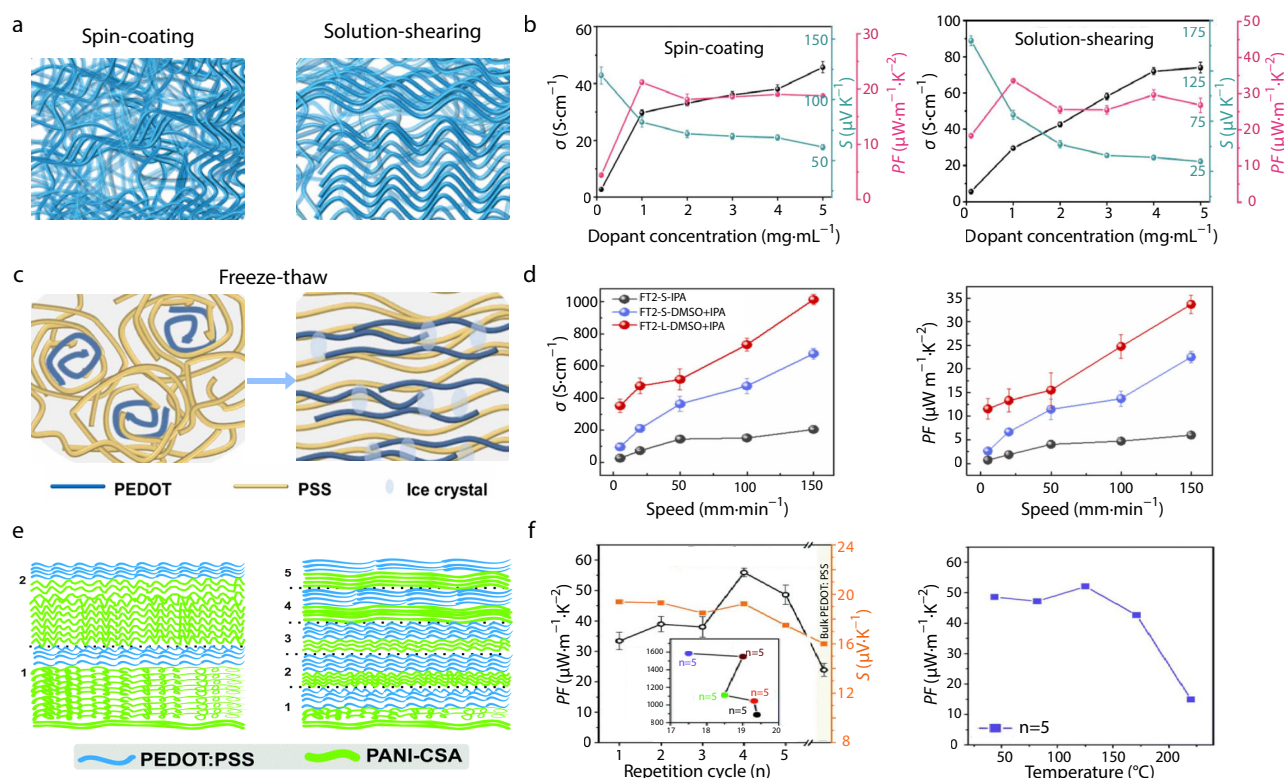


Fig. 6 (a) Schematics of aggregate states in polymer films fabricated by spin coating and solution shearing; (b) TE performance of doped films fabricated via spin-coating (left) and solution-shearing (right); (a, b: reproduced with permission from Ref. [34]; Copyright (2023), Royal Society of Chemistry.) (c) Schematic illustrating the structural evolution by freeze-thaw treatments; (d) The σ and PF of gel fibers; (c, d: reproduced with permission from Ref. [75]; Copyright (2022), Elsevier.) (e) Schematic explaining the stretching of the chains of both the PEDOT:PSS and PANI-CSA layer with increasing repetition cycles n ; (f) PF and S as functions of repetition cycles (n) in the PEDOT: PSS/PANI-CSA multilayer structure, with the corresponding values for bulk PEDOT:PSS shown for comparison. The inset shows the variation of σ and the S with the repetition cycle (left). The temperature-dependent PF of multilayer films for the multilayer film with $n=5$ (right). (e, f: reproduced with permission from Ref. [76]; Copyright (2016), Royal Society of Chemistry.)

air-material interfaces, which effectively modulate phonon transport and induce intense phonon scattering. Leveraging porous architectures to reduce κ has emerged as a key strategy for optimizing the TE performance. Porous structures represent an effective approach for reducing κ of polymers. Weinbach *et al.* provided a detailed review of porous materials and κ .^[79] Zhao *et al.* achieved reduced κ by interface scattering in porous films.^[25] Chen *et al.* prepared aerogels by freeze-drying a composite of cellulose nanofibers and polyaniline solution, achieved a reduction in κ to $0.085 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$.^[80] Engineering a superlattice architecture that functions as a phonon low-pass and electron high-pass filter is a key method for impeding lattice thermal transport. This effect is twofold: first, interfacial scattering shortens the phonon mean free path; second, the periodicity of the film structure induces Brillouin zone folding, which suppresses the phonon group velocity and consequently lowers the lattice thermal conductivity (κ_L). Wang *et al.* proposed a method for constructing PMHJ structure (Fig. 7a) that successfully represented the κ of the film.^[8] The PMHJ assembly was fabricated using the organic semiconductors PDPPSe-12 and PBTTT, with a cross-linking agent (4Bx) introduced to enable periodic layering (Figs. 7b and 7c). A systematic investigation revealed that the resulting film comprised alternating polymer

layers and distinct interface layers, the latter exhibiting characteristics analogous to those of bulk heterojunctions. A key finding of this study is that when the thickness of the individual polymer layers approaches the average phonon mean free path along the conjugated backbone or the π - π stacking direction, the size effect strongly enhances interface scattering. As a result, the reducing individual layer thickness further suppressed in-plane thermal conductivity ($\kappa_{//}$), with the (6, 4, 4) configuration achieving $0.1 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ (Fig. 7d), and the κ_L of PDPPSe-12 and PBTTT films was significantly reduced by 55% and 76%, respectively (Fig. 7e). This study underscores the unique capability of the PMHJ architecture to minimize the κ_L of organic materials. Ultimately, the doped films exhibited a low κ of $0.18 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ and a maximum ZT value of 1.28. In addition, the method is compatible with large-area solution processing techniques, enabling the fabrication of integrated TE devices with outstanding normalized power density (Fig. 7f). In summary, the PMHJ architecture offers a novel strategy for the development of high-performance OTE materials.

Decoupling of the Seebeck Coefficient and Electrical Conductivity

A challenge in advancing OTE lies in the inherent trade-off between σ and S . Typically, increasing the carrier concentration to

enhance σ tends to suppress S . To overcome this compromise and achieve effective decoupling of σ and S , two primary approaches are currently pursued: improving the μ in the doped state and partially alleviating the constraint between σ and S . This is achieved through strategies such as optimizing the molecular structures and innovating doping methods, which aim to minimize the impact of dopants on the aggregated structure of the material. Consequently, the carrier delocalization length increases, ultimately leading to a higher μ . On the other hand, optimizing molecular orientation provides an important pathway for simultaneously enhancing both σ and S . By tailoring the polymer molecular orientation through processing techniques, the charge transport pathways, electronic structure, and intermolecular interactions in polymers can be directly in-

fluenced, thereby regulating key parameters determining TE performance.^[81]

We will focus on controlling the molecular orientation of polymer using processing techniques. For example, Wang *et al.* fabricated one diketopyrrolopyrrole (DPP)-based polymer (DPP-BTz) films with varying molecular orientations using a dynamic template-assisted solution shearing method (Fig. 8a).^[24] The structural characteristics of these three types of films were subsequently investigated using grazing incidence wide-angle X-ray scattering (GIWAXS) (Fig. 8b). The results revealed that increasing the lamellar spacing in the bimodal film could reduce the diffusion barrier of dopants and enhance the doping level. Moreover, the bimodal film exhibited a higher energy level difference between the Fermi level

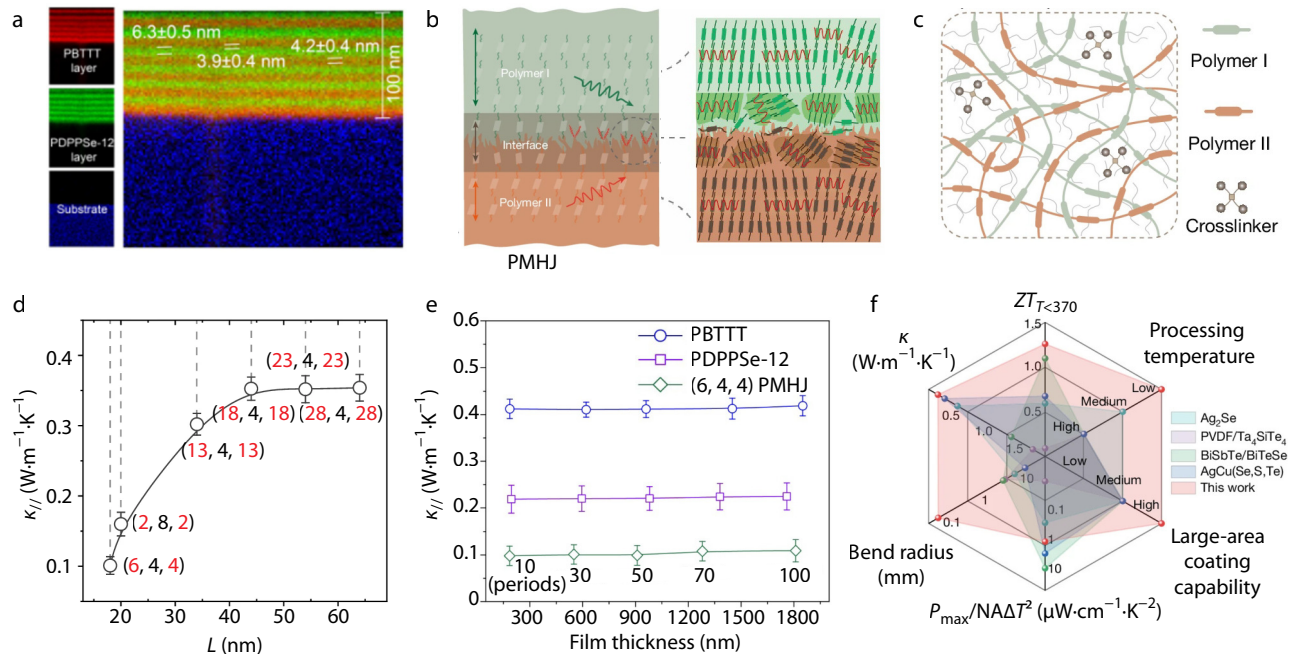
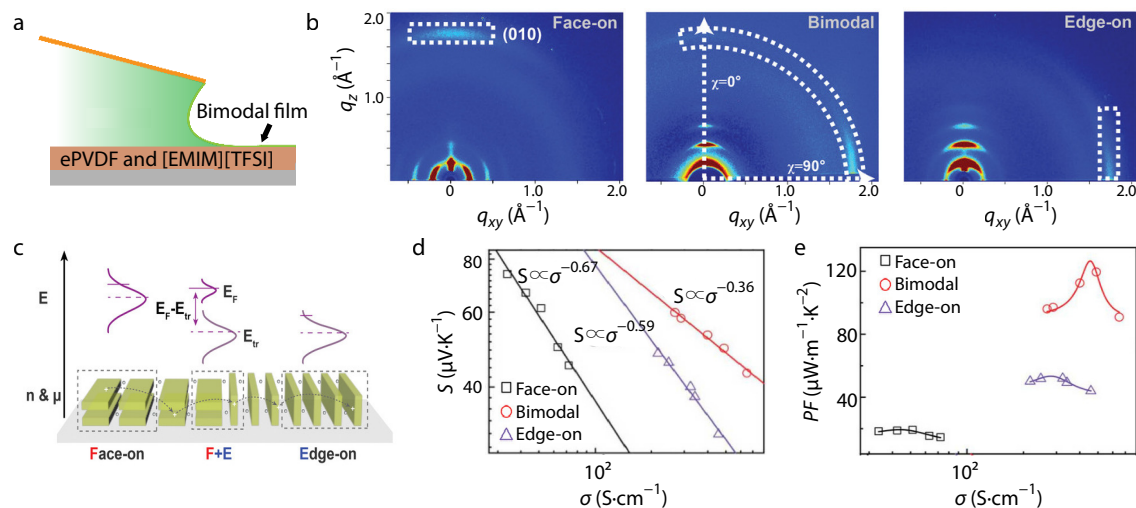


Fig. 7 (a) Time of flight secondary ion mass spectrometry cross-sectional images along y-z axis of a PDPPSe-12:PBTtT PMHJ (six period) film; (b) Schematic illustration of PMHJ geometry; (c) PDPPSe-12 and PBTtT were photo-crosslinked by crosslinking agent; (d) Period-thickness-dependent $\kappa_{||}$ of undoped PDPPSe-12: PBTtT PMHJ film; (e) The $\kappa_{||}$ of undoped PBTtT, PDPPSe-12 and (6,4,4) PMHJ films; (f) Radar charts for comparing the performance of typical flexible TE films. (Reproduced with permission from Ref. [8]; Copyright (2024), Springer Nature.)



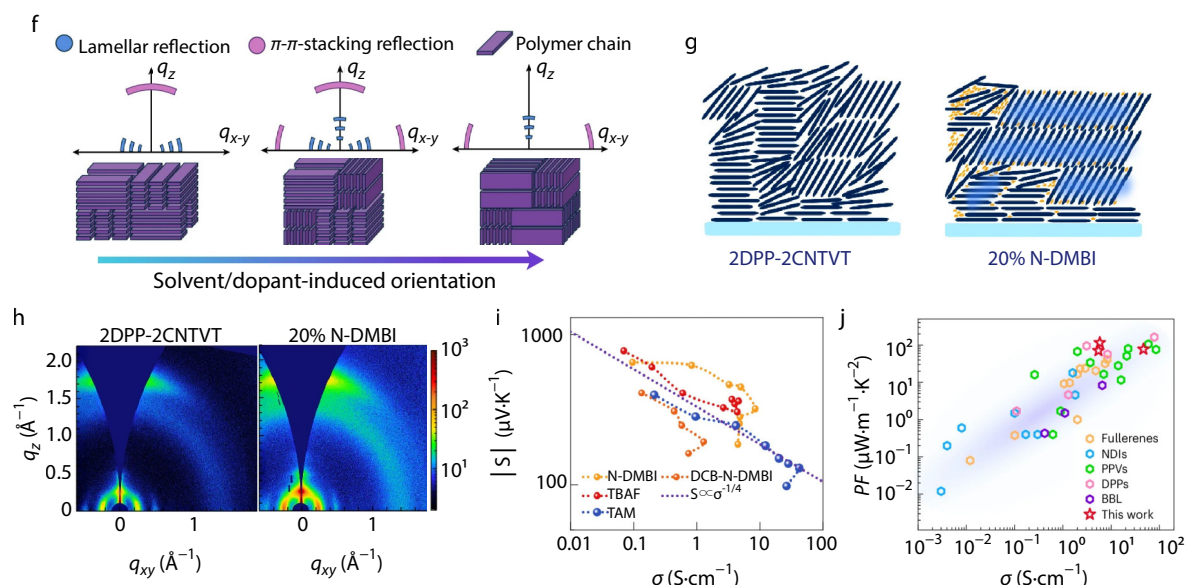


Fig. 8 (a) The fabrication process of bimodal films; (b) GIXRD 2D patterns of DPP-BTz films at face-on, bimodal, and edge-on orientations; (c) Schematic of the correlation between molecular orientations; (d) The S versus σ relation at room temperature, comparing films with face-on/edge-on and bimodal molecular orientations; (e) The PF versus σ relation at room temperature; (a–e: reproduced with permission from Ref. [24]; Copyright (2022), Wiley.) (f) Schematic of the MFDA process in which the solution aggregates modulate the molecular orientation of the film; (g) Schematic depicting the microscale morphology of pristine film and the film after doping; (h) GIWAXS patterns of 2DPP-2CNTVT film and the 2DPP-2CNTVT film after doping; (i) TE properties of sequentially doped 2DPP-2CNTVT; (j) PF versus σ of previously reported n-type OTEs. (f–j: reproduced with permission from Ref. [82]; Copyright (2025), Springer Nature.)

(E_F) and the transmission level (E_{tr}), resulting in improved σ and S (Fig. 8c). Compared to the face-on and edge-on films, the bimodal film exhibits a weakened trade-off between σ and S (Fig. 8d). The bimodal film achieved a PF as high as $346 \mu\text{W}\cdot\text{m}^{-1}\cdot\text{K}^{-2}$ (Fig. 8e) and a ZT value of 0.40, representing a fourfold improvement in the TE performance compared to films with a single molecular orientation. Villalva *et al.* proposed a molecular-force-driven anisotropy (MFDA) method for regulating polymer orientation (Fig. 8f).^[82] Based on the Hansen solubility parameter (HSP) framework, solvent selection criteria were established to enable conjugated polymers in membranes to achieve preferential orientation. Furthermore, incorporating 4-(2,3-dihydro-1,3-dimethyl-1H-benzimidazol-2-yl)-*N,N*-dimethylbenzenamine (N-DMBI) as a dopant also enhanced the polymer crystallinity and reduced the structural defects through favorable intermolecular interactions (Fig. 8g). The edge-on orientation significantly increases the in-plane charge delocalization length (Fig. 8h), leading to a simultaneous enhancement in both σ and S , increasing a PF of $115 \mu\text{W}\cdot\text{m}^{-1}\cdot\text{K}^{-2}$ (Figs. 8i and 8j). This synergistic effect effectively overcomes the conventional trade-off in TE performance.

CONCLUSIONS AND PERSPECTIVES

Polymer processing technology serves as a critical bridge for the practical application of OTE materials. This review systematically outlines key advancements in the field in recent years, spanning solution-phase deposition techniques that regulate molecular orientation and crystallinity, patterning methods enabling precise micro-/nano-structural construction, and large-area fabrication processes aimed at scalable production. These process-

ing strategies effectively enhance the σ and S while reducing the κ through precise control over the polymer microstructures, thereby driving continuous breakthroughs in performance.

To realize the commercialization and unlock the full potential of OTE materials, the following directions are crucial.

Current research has demonstrated that strategies such as orientation engineering and heterostructure construction can synergistically improve σ and S to some extent or reduce κ without significantly compromising the electrical properties. The core future challenge lies in fundamentally decoupling the strong interdependencies among σ , S , and κ . This requires synergistic innovation at both the molecular design and processing engineering levels: developing novel molecular structures to provide more flexible tuning dimensions for these parameters, and creating more sophisticated processing methods (*e.g.*, multi-scale structural) to achieve independent and precise control over charge and phonon transport pathways, ultimately approaching the ideal "phonon glass, electron crystal" state.

High-performance laboratory results must be translated into practical products via stable, reliable, and low-cost manufacturing processes. Future development in processing technologies should focus on high throughput and automation. The optimization and integration of large-area, continuous production techniques, such as roll-to-roll printing and high-speed spraying, are key. More importantly, the introduction of machine learning and artificial intelligence technologies, through the modeling and analysis of vast amounts of process-structure-performance data, can enable intelligent optimization of processing parameters, inverse design of microstructures, and online monitoring and defect prediction

during production. This will significantly improve the manufacturing efficiency, reproducibility, and yield, paving the way for fully automated production lines.

The advantages of OTE materials, such as flexibility, solution processability, and biocompatibility, render them highly promising for multifunctional integrated systems. Future devices should evolve towards modular systems that integrate power generation, cooling, and sensing. For example, the TE effect can be leveraged to simultaneously achieve energy harvesting from body temperature differences (power generation), precise local temperature management (cooling), and pressure/heat flow sensing. Such intelligent, self-powered, flexible integrated modules will find applications in areas such as wearable electronic health monitoring, Internet of Things (IoT) sensing, and low-power flexible storage systems.

In summary, by continuously advancing the integration of high-performance materials with precision processing technologies, and actively exploring cross-innovation with intelligent manufacturing and multifunctional system integration, OTE technology holds the potential to open up entirely new application fields in the future energy and electronic information sectors.

BIOGRAPHIES

Li-Yao Liu received her Ph.D. degree from Institute of Chemistry, Chinese Academy of Sciences (ICCAS). She joined the CAS Key Laboratory of Organic Solids, ICCAS in 2019. She mainly engaged in research on organic thermoelectric materials and devices.

Chong-An Di is a professor of the Organic Solids Laboratory in ICCAS. He finished his Ph.D. study in chemistry from ICCAS in 2008 and was appointed as a professor in 2016 at ICCAS. Currently, his research focuses on the investigation of organic thin film transistors and organic thermoelectrics.

Conflict of Interests

The authors declare no interest conflict.

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REFERENCES

- Shi, X. L.; Zou, J.; Chen, Z. G. Advanced thermoelectric design: from materials and structures to devices. *Chem. Rev.* **2020**, *120*, 7399–7515.
- Ma, Z. L.; Ma, Z. Z.; Zhao, L. M.; Zhang, J. L.; Guo, P. F.; Li, Q.; Wang, H. Q. Integration and application of solar-responsive energy storage systems. *Adv. Energy Mater.* **2025**, *15*, e03128.
- Cao, J. L.; Chen, Z. X.; Kong, S.; Liu, L.; Wang, R. X. Towards urban wind utilization: the spatial characteristics of wind energy in urban areas. *J. Clean. Prod.* **2024**, *450*, 141981.
- Zhang, X. W.; Ji, R. W.; Sun, K.; Zhang, J. H.; Zhang, X. J.; Yin, M. W.; Kong, M.; Reabroy, R. A review of ocean tidal current energy technology: advances, trends, and challenges. *Phys. Fluids.* **2025**, *37*, 071308.
- Ji, Z.; Li, Z. Y.; Liu, L. Y.; Zou, Y.; Di, C. A.; Zhu, D. B. Organic thermoelectric devices for energy harvesting and sensing applications. *Adv. Mater. Technol.* **2024**, *9*, 2302128.
- Zhao, Y.; Liu, L. Y.; Zhang, F. J.; Di, C. A.; Zhu, D. B. Advances in organic thermoelectric materials and devices for smart applications. *SmartMat* **2021**, *2*, 426–445.
- Tang, H. R.; Liang, Y. Y.; Liu, C. C.; Hu, Z. C.; Deng, Y. F.; Guo, H.; Yu, Z. D.; Song, A.; Zhao, H. Y.; Zhao, D. K.; Zhang, Y. Z.; Guo, X. G.; Pei, J.; Ma, Y. G.; Cao, Y.; Huang, F. A solution-processed n-type conducting polymer with ultrahigh conductivity. *Nature* **2022**, *611*, 271–277.
- Wang, D. Y.; Ding, J. M.; Ma, Y. Q.; Xu, C. L.; Li, Z. Y.; Zhang, X.; Zhao, Y.; Zhao, Y.; Di, Y. Q.; Liu, L. Y.; Dai, X. J.; Zou, Y.; Kim, B. S.; Zhang, F. J.; Liu, Z. T.; McCulloch, I.; Lee, M.; Chang, C.; Yang, X.; Wang, D.; Zhang, D. Q.; Zhao, L. D.; Di, C. A.; Zhu, D. B. Multi-heterojunctioned plastics with high thermoelectric figure of merit. *Nature* **2024**, *632*, 528–535.
- Zhao, Y. R.; Li, Y. M.; Yang, R. H.; Song, Z. Y.; Sun, M. Y.; Fan, N.; Zhong, W. K.; Ye, L.; Yuan, Y.; Jiao, F.; Wang, Y. X.; Hu, W. P. Environmentally stable n-type conducting polymer with high intrinsic stretchability. *Adv. Mater.* **2026**, DOI: 10.1002/adma.202518190.
- Liu, K.; Wang, J. Y.; Pan, X. R.; Tian, S. Y.; Liu, Y. D.; Zhang, Z.; Di, Y. Q.; Chen, J. P.; Wu, C. W.; Deng, C. Y.; Wang, D. Y.; Li, P. Y.; Pan, C. K.; Qi, F. L.; Liu, J. H.; Hua, J.; Pei, J.; Di, C. A.; Guo, Y. L.; Liu, Y. Q.; Lei, T. N-type thermoelectric elastomers. *Nature* **2025**, *644*, 920–926.
- Jo, G.; Jeong, W. J.; Choi, S.; Kim, H.; Park, J. J.; Jung, J.; Chang, M. Large-scale alignment of polymer semiconductor nanowires for efficient charge transport via controlled evaporation of confined fluids. *ACS APPL. Mater. Interfaces* **2019**, *11*, 1135–1142.
- Zhang, X.; Ding, J. M.; Wang, D. Y.; Dai, X. J.; Ma, Y. Q.; Zhao, Y.; Guan, B.; Liu, L. Y.; Zou, Y.; Zhang, F. J.; Di, C. A. Waterborne paints based on polymeric semiconductor for attachable thermoelectric generators. *Small Struct.* **2022**, *4*, 2200278.
- Zhao, W. R.; Ding, J. M.; Zou, Y.; Di, C. A.; Zhu, D. B. Chemical doping of organic semiconductors for thermoelectric applications. *Chem. Soc. Rev.* **2020**, *49*, 7210–7228.
- Song, Y. L.; Ding, J. M.; Dai, X. J.; Li, C.; Di, C. A.; Zhang, D. Q. Enhancement of the thermoelectric performance of n-type naphthalene diimide-based conjugated polymer by engineering of side alkyl chains. *ACS Mater. Lett.* **2022**, *4*, 7210–7228.
- Ding, J. M.; Liu, Z. T.; Zhao, W. R.; Jin, W. L.; Xiang, L. Y.; Wang, Z. J.; Zeng, Y.; Zou, Y.; Zhang, F. J.; Yi, Y. P.; Diao, Y.; McNeill, C. R.; Di, C. A.; Zhang, D. Q.; Zhu, D. B. Selenium-substituted diketopyrrolopyrrole polymer for high-performance p-type organic thermoelectric materials. *Angew. Chem. Int. Ed.* **2019**, *58*, 19170–19175.
- Shirakawa, H.; Louis, E. J.; Macdiarmid, A. G.; Chiang, C. K.; Heeger, A. J. Synthesis of electrically conducting organic polymers: halogen derivatives of polyacetylene, (CH)_x. *J. C. S. Chem. Comm.* **1977**, 578–580.
- Reghu, M.; Cao, Y.; Moses, D.; Heeger, A. J. Counterion-induced processibility of polyaniline: transport at the metal-insulator boundary. *Phys. Rev. B* **1993**, *47*, 1758–1764.
- Bubnova, O.; Khan, Z. U.; Malti, A.; Braun, S.; Fahlman, M.; Berggren, M.; Crispin, X. Optimization of the thermoelectric figure of merit in the conducting polymer poly(3, 4-ethylenedioxythiophene). *Nat. Mater.* **2011**, *10*, 429–733.
- Søndergaard, R. R.; Hösel, M.; Espinosa, N.; Jørgensen, M.; Krebs,

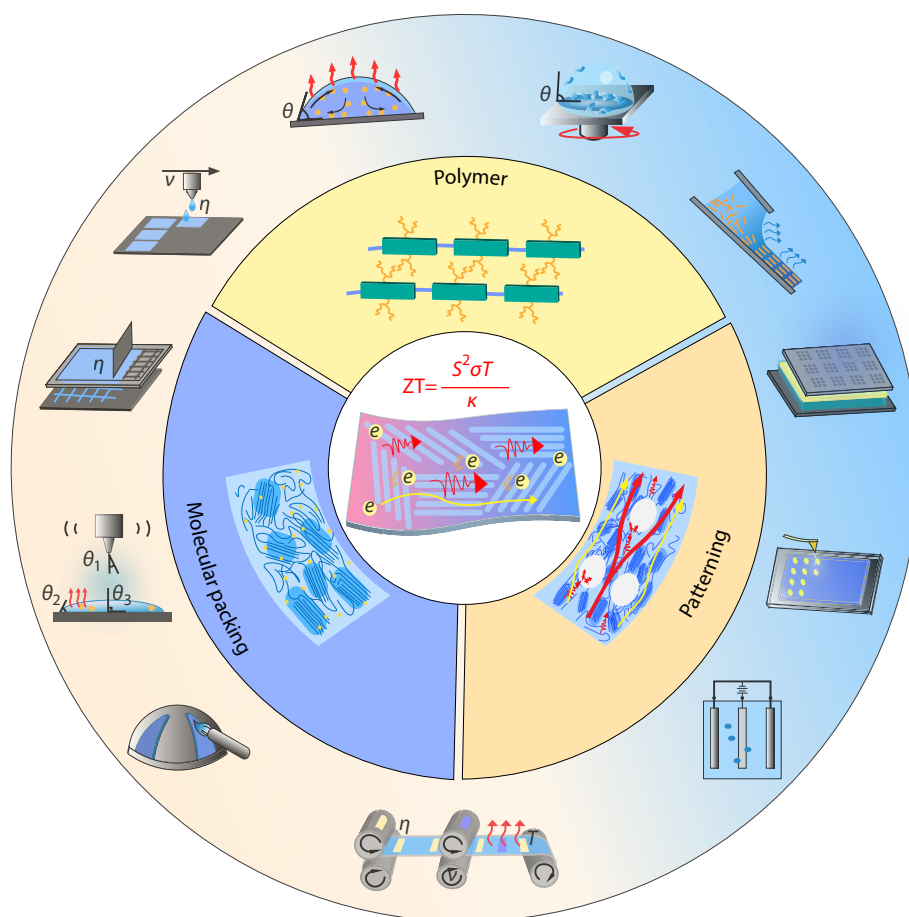
Graphical Abstract

Engineering Process of Polymer for Thermoelectric Applications

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This review systematically summarizes the latest advances in polymer processing techniques for thermoelectric applications and explores the intrinsic relationships among the processing methods, microstructure, and thermoelectric performance. Finally, future perspectives are presented for the industrialization of high-performance organic thermoelectric devices.



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- F. C. Practical evaluation of organic polymer thermoelectrics by large-area R2R processing on flexible substrates. *Energy Sci. Eng.* **2013**, 1, 81–88.
- 20 Hong, C. T.; Kang, Y. H.; Ryu, J.; Cho, S. Y.; Jang, K. S. Spray-printed CNT/P3HT organic thermoelectric films and power generators. *J. Mater. Chem. A* **2015**, 3, 21428.
- 21 Sun, Y. H.; Qiu, L.; Tang, L. P.; Geng, H.; Wang, H. F.; Zhang, F. J.; Huang, D. Z.; Xu, W.; Yue, P.; Guan, Y. S.; Jiao, F.; Sun, Y. M.; Tang, D. W.; Di, D. A.; Yi, Y. P.; Zhu, D. B. Flexible n-type high-performance thermoelectric thin films of poly(nickel-ethylenetetra-thiolate) prepared by an electrochemical method. *Adv. Mater.* **2016**, 28, 3351–3358.
- 22 Hamidi-Sakr, A.; Biniek, L.; Bantignies, J. L.; Maurin, D.; Herrmann, L.; Leclerc, N.; L  v  que, P.; Vijayakumar, V.; Zimmermann, N.; Brinkmann, M. A versatile method to fabricate highly in-plane aligned conducting polymer films with anisotropic charge transport and thermoelectric properties: the key role of alkyl side chain layers on the doping mechanism. *Adv. Funct. Mater.* **2017**, 27, 1700173.
- 23 Brus, V. V.; Gluba, M.; Rappich, J.; Lang, F.; Maryanchuk, P. D.;

- Nickel, N. H. Fine art of thermoelectricity. *ACS Appl. Mater. Interfaces* **2018**, *10*, 4737–4742.
- 24 Wang, D. Y.; Ding, J. M.; Dai, X. J.; Xiang, L. Y.; Ye, D. L.; He, Z. H.; Zhang, F. J.; Jung, S. H.; Lee, J. K.; Di, C. A.; Zhu, D. B. Triggering ZT to 0.40 by engineering orientation in one polymeric semiconductor. *Adv. Mater.* **2022**, *35*, 2208215.
 - 25 Zhao, Y.; Li, Z. Y.; Wang, D. Y.; Zhang, X.; Ji, Z.; Niu, L. X.; Di, Y. Q.; Guan, Y.; Liu, L. Y.; Zou, Y.; Li, C.; Zhang, F. J.; Zhang, D. Q.; Zhu, D. B.; Di, C. A. High performance and colorful polymer thermoelectrics with imprinted porous film. *Adv. Mater.* **2024**, *36*, 2407692.
 - 26 Wang, X. Y.; Ding, Y. F.; Zhang, X. Y.; Zhou, Y. Y.; Pan, C. K.; Li, Y. J.; Liu, N. F.; Yao, Z. F.; Chen, Y. S.; Xie, Z. H.; Huang, Y. F.; Xu, Y. F.; Wu, H. T.; Huang, C. X.; Xiong, M.; Ding, L.; Yu, Z. D.; Li, Q. Y.; Zheng, Y. Q.; Wang, J. Y.; Pei, J. Light-triggered regionally controlled n-doping of organic semiconductors. *Nature* **2025**, *642*, 599–605.
 - 27 Diao, Y.; Shaw, L.; Bao, Z. N.; Mannsfeld, S. C. B. Morphology control strategies for solution-processed organic semiconductor thin films. *Energy Environ. Sci.* **2014**, *7*, 2145–2159.
 - 28 Huq, A. F.; Ammar, A.; Al-Enizi A. M.; Karim, A. In-situ orientation and crystal growth kinetics of P3HT in drop cast P3HT: PCBM films. *Polymer* **2017**, *113*, 200–213.
 - 29 Chen, L.; Chi, S. J.; Zhao, K. F.; Liu, J. G.; Yu, X. H.; Han, Y. C. Aligned films of the DPP-based conjugated polymer by solvent vapor enhanced drop casting. *Polymer* **2016**, *104*, 123–129.
 - 30 Li, H.; Liang, Y.; Liu, Y. L.; Liu, S. Q.; Li, P. C.; He, C. B. Engineering doping level for enhanced thermoelectric performance of carbon nanotubes/polyaniline composites. *Compos. Sci. Technol.* **2021**, *210*, 108797.
 - 31 Li, H.; Liu, Y. L.; Li, P. C.; Liu, S. Q.; Du, F. P.; He, C. B. Enhanced thermoelectric performance of carbon nanotubes/polyaniline composites by multiple interface engineering. *ACS Appl. Mater. Interfaces* **2021**, *13*, 6650–6658.
 - 32 Yoo, D.; Son, W. Y.; Kim, S. Y.; Lee, J. J.; Lee, S. H.; Choi H. H.; Kim, J. H. Gradual thickness-dependent enhancement of the thermoelectric properties of PEDOT:PSS nanofilms. *RSC Adv.* **2014**, *4*, 58924–58929.
 - 33 Zhang, F. J.; Di, C. A.; Berdonov, N.; Hu, Y. T.; Hu, Y. B.; Gao, X. K.; Meng, Q.; Sirringhaus H.; Zhu, D. B. Ultrathin film organic transistors: precise control of semiconductor thickness via spin-coating. *Adv. Mater.* **2012**, *25*, 1401–1407.
 - 34 Feng, K.; Yang, W. L.; Jeong, S. Y.; Ma, S. X.; Li, Y. C.; Wang, J. W.; Wang, Y. M.; Woo, H. Y.; Chan, P. K. L.; Wang, G.; Guo, X. G.; Zhu, M. F. Cyano-functionalized fused bithiophene imide dimer-based n-type polymers for high-performance organic thermoelectrics. *Adv. Mater.* **2023**, *35*, 2408716.
 - 35 Xu, J.; Wu, H. C.; Zhu, C. X.; Ehrlich, A.; Shaw, L.; Nikolka, M.; Wang, S. H.; Molina-Lopez, F.; Gu, X. D.; Luo, S. C.; Zhou, D. S.; Kim, Y. H.; Wang, G. J. N.; Gu, K.; Feig, V. R.; Chen, S. C.; Kim, Y.; Katsumata, T.; Zheng, Y. Q.; Yan, H.; Chung, J. W.; Lopez, J.; Murmann, B.; Bao, Z. N. Multi-scale ordering in highly stretchable polymer semiconducting films. *Nat. Mater.* **2019**, *18*, 594–601.
 - 36 Yang, X. F.; Sun, M. L.; Du, L.; Wu, F. Y.; Kim, H. H.; Woo, H. Y. L.; Chen, L.; Chen, Y. W. Co-solvent strategy enables high-performance printing organic thermoelectric via synergistic modulation of crystallinity and film homogeneity. *Small* **2025**, *21*, e10495.
 - 37 Perry, S. C.; Akbar, S.; Clarke, R.; Shah, S. Z. H.; Nandhakumar, I. Electrodeposition of thermoelectric materials. *Chem. Electro. Chem.* **2025**, *12*, e202500052.
 - 38 Teixeira, H. J.; Dias, C.; Veloso, R. C.; Apolinário A.; Ventura, J. Tuning PEDOT: PSS low-impedance thin films with high charge injection for microelectrodes applications. *Prog. Org. Coat.* **2022**, *168*, 106894.
 - 39 Su, L. S.; Gan, Y. X. Experimental study on synthesizing TiO₂ nanotube/polyaniline (PANI) nanocomposites and their thermoelectric and photosensitive property characterization. *Compos. Part B: Eng.* **2012**, *43*, 170–182.
 - 40 Culebras, M.; Igual-Muñoz, A. M.; Rodríguez-Fernández, C.; Gómez-Gómez, M. I.; Gómez C.; Cantarero, A. Manufacturing Te/PEDOT films for thermoelectric applications. *ACS Appl. Mater. Interfaces* **2017**, *9*, 20826–20832.
 - 41 Li, Y.; Chen, Z. P.; Gao, C. Y.; Li, H. P.; Fan, X. H.; Cao, X. B.; Yang, L. M. Electrochemical assembly of flexible PPy/Bi-Te alloy/PPy thermoelectric composite films with a sandwich-type structure. *New J. Chem.* **2024**, *48*, 1634–1641.
 - 42 Liu, L. Y.; Sun, Y. H.; Li, W. B.; Zhang, J. J.; Huang, X.; Chen, Z. J.; Sun, Y. M.; Di, C. A.; Xu, W.; Zhu, D. B. Flexible unipolar thermoelectric devices based on patterned poly[K_x(Ni-ethylenetetra-thiolate)] thin films. *Mater. Chem. Front.* **2017**, *1*, 2111–2116.
 - 43 Codau, T. C.; Codau, E. Synthesis of ultra-stretchable thermoelectric nanofibrous membrane based on wet-electrospun polyurethane/MWCNTs composites. *Mater. Today Sustain.* **2024**, *27*, 100831.
 - 44 Liu, H. R.; Liu, D.; Yang, J. C.; Gao, H. F.; Wu, Y. C. Flexible electronics based on organic semiconductors: from patterned assembly to integrated applications. *Small* **2023**, *19*, 2206938.
 - 45 Li, Y.; Chen, R. Z. Photolithographic patterning of organic semiconductors for organic field-effect transistors. *Chem. Rev.* **2025**, *125*, 4933–4973.
 - 46 Qiu, M. J.; Du, W. W.; Zhou, S. Y.; Cai, P. Z.; Luo, Y. W.; Wang, X. X.; Yang R.; Zhao, J. J. Recent progress in non-photolithographic patterning of polymer thin films. *Prog. Polym. Sci.* **2023**, *142*, 101688.
 - 47 Choi, W.; Kim, Y. E.; Yoo, H. Patterning of organic semiconductors leads to functional integration: from unit device to integrated electronics. *Polymers* **2024**, *16*, 2613.
 - 48 Park, T.; Kim, M.; Lee, E. K.; Hur, J.; Yoo, H. Overcoming downscaling limitations in organic semiconductors: strategies and progress. *Small* **2023**, *20*, 2306468.
 - 49 Ji, Z.; Li, Z. Y.; Dai, X. J.; Xiang, L. Y.; Zhao, Y.; Wang, D. Y.; Zhang, X.; Liu, L. Y.; Han, Z. Y.; Niu, L. X.; Di, Y. Q.; Zou, Y.; Di, C. A.; Zhu, D. B. Organic photoexcitation-assisted molecular doping for high-performance polymeric thermoelectric materials. *JACS Au* **2024**, *4*, 3884–3895.
 - 50 Xue, X.; Li, C.; Zhou, Q.; Yu, X. B.; Gao, C. Y.; Chenchai, H. Y.; Liao, J. C.; Shangguan, Z. C.; Zhang, X. S.; Zhang, G. X.; Zhang, D. Q. Conjugated polymer-based photo-crosslinker for efficient photo-patterning of polymer semiconductors. *Adv. Mater.* **2024**, *36*, 2407305.
 - 51 Zhu, X. C.; Ekinici, H.; Pan, A.; Cui, B.; Zhu, X. L. Electron beam lithography on nonplanar and irregular surfaces. *Microsyst. Nanoeng.* **2024**, *10*, 52.
 - 52 Lee, H.; Lee, S.; Cha, S.; Anoop, G.; Shin, H. Nanoscale electron beam patterning of PEDOT: PSS free-standing films for enhanced thermoelectric performance. *Energy Environ. Mater.* **2024**, *8*, e12824.
 - 53 Xiang, L. Y.; He, Z. H.; Yan, C. Y.; Zhao, Y.; Li, Z. Y.; Jia, L. X.; Jiang, Z. L.; Dai, X. J.; Lemaury, V.; Ma, Y. Q.; Liu, L. Y.; Meng, Q.; Zou, Y.; Beljonne, D.; Zhang, F. J.; Zhang, D. Q.; Di, C. A.; Zhu, D. B. Nanoscale doping of polymeric semiconductors with confined electrochemical ion implantation. *Nat. Nanotechnol.* **2024**, *19*, 1122–1129.
 - 54 Choudhary, K.; Chen, A. X.; Pitch, G. M.; Runser, R.; Urbina, A.; Dunn, T.; Kodur, M.; Kleinschmidt, A.; Wang, B.; Bunch, J.; Fenning, D.; Ayzner, A.; Lipomi, D.; Comparison of the mechanical properties of a conjugated polymer deposited using spin coating, interfacial spreading, solution shearing, and spray

- coating. *ACS Appl. Mater. Interfaces* **2021**, *13*, 51436–51446.
- 55 Ren, W. B.; Yang, M. Z.; Zhou, L.; Fan, Y. J.; He, S.; Pan, J. Y.; Tang, T. X.; Xiao, Y.; Nan, C. W.; Shen, Y. Scalable ultrathin all-organic polymer dielectric films for high-temperature capacitive energy storage. *Adv. Mater.* **2022**, *34*, 2207421.
 - 56 Lin, Y. T.; Lee, C. Y.; Wu, C. Y.; Lin, J. M.; Lee, T. C.; Tung, S. H.; Liu, C. L. High thermoelectric performance of spray-coated poly(3, 4-ethylenedioxythiophene): poly (styrenesulfonate) films enabled by two-step post-treatment process. *J. Power Sources* **2023**, *556*, 232516.
 - 57 Darabi, S.; Yang, C. Y.; Li, Z. R.; Huang, J. D.; Hummel, M.; Sixta, H.; Fabiano, S.; Müller, C. Polymer-based n-type yarn for organic thermoelectric textiles. *Adv. Electron. Mater.* **2023**, *9*, 2201235.
 - 58 Sun, T. T.; Zhou, B. Y.; Zheng, Q.; Wang, L. J.; Jiang, W.; Snyder, G. J. Stretchable fabric generates electric power from woven thermoelectric fibers. *Nat. Commun.* **2020**, *11*, 572.
 - 59 Chang, T. H.; Akin, S.; Cho, S. S.; Lee, J. S.; Lee, S. A.; Park, T. W.; Hong, S. K.; Yu, T. H.; Ji, Y. Y.; Yi, J. H.; Gong, L. S.; Kim, D. R.; Kim, Y. L.; Jun, M. B.; Lee, C. H. In situ spray polymerization of conductive polymers for personalized e-textiles. *ACS Nano* **2023**, *17*, 22733–22743.
 - 60 Hossain, M. S.; Li, T. Z.; Yu, Y.; Yong, J.; Bahk, J. H.; Skafidas, E. Recent advances in printable thermoelectric devices: materials, printing techniques, and applications. *RSC Adv.* **2020**, *10*, 8421–8434.
 - 61 Ferhat, S.; Domain, C.; Vidal, J.; Noël, D.; Ratiera, B.; Lucas, B. Organic thermoelectric devices based on a stable n-type nanocomposite printed on paper. *Sustain. Energ. Fuels* **2018**, *2*, 199–208.
 - 62 Jing, J.; Chopplet, L.; Battaglini, N.; Noël, V.; Piro, B.; Leydecker, T.; Wang, Z. M.; Mattana, G.; Orgiu, E. The role of substrates and electrodes in inkjet-printed PEDOT: PSS thermoelectric generators. *J. Mater. Chem. A* **2024**, *12*, 6185–6192.
 - 63 Zheng, C. Z.; Xiang, L. Y.; Jin, W. L.; Shen, H. G.; Zhao, W. R.; Zhang, F. J.; Di, C. A.; Zhu, D. B. A flexible self-powered sensing element with integrated organic thermoelectric generator. *Adv. Mater. Technol.* **2019**, *4*, 1900247.
 - 64 Wei, Q. S.; Mukaida, M.; Kirihaara, K.; Naitoh, Y.; Ishida, T. Polymer thermoelectric modules screen-printed on paper. *RSC Adv.* **2014**, *4*, 28802–28806.
 - 65 He, X.; He, R. H.; Lan, Q. M.; Wu, W. J.; Duan, F.; Xiao, J. D.; Zhang, M.; Zeng, Q. G.; Wu, J. H.; Liu, J. Y. Screen-printed fabrication of PEDOT: PSS/Silver nanowire composite films for transparent heaters. *Materials* **2017**, *10*, 200.
 - 66 Nayak, R.; Shetty, P.; Selvakumar, M.; Rao, A.; Sriram K. V.; Wagle, S.; Nayak, S.; Kamath, V.; Shetty, N.; Saquib, M. Formulation and optimization of copper selenide/PANI hybrid screen printing ink for enhancing the power factor of flexible thermoelectric generator: a synergetic approach. *Ceram. Int.* **2024**, *50*, 25779–25791.
 - 67 Zhang, B.; Yu, T. B.; Meulemans, A.; Naenen, V.; Rosenthal, M.; Escudero, D.; Molina-Lopez, F. Brush and dry: a posttreatment-free simple process to pattern PEDOT: PSS films with enhanced thermoelectric performance via shear flow-induced morphology tuning. *ACS Appl. Mater. Interfaces* **2025**, *17*, 20236–20247.
 - 68 Zhang, Z. M.; Qiu, J. J.; Wang, S. R. Roll-to-roll printing of flexible thin-film organic thermoelectrics devices. *Manuf. Lett.* **2016**, *8*, 6–10.
 - 69 Zhang, Z. M.; Wang, B. R.; Qiu, J. J.; Wang, S. R. Roll-to-roll printing of spatial wearable thermoelectrics. *Manuf. Lett.* **2019**, *21*, 28–34.
 - 70 Zhang, Z.; Li, P. Y.; Xiong, M.; Zhang, L.; Chen, J. P.; Lei, X.; Pan, X. R.; Wang, X.; Deng, X. Y.; Shen, W. Y.; Mei, Z.; Liu, K. K.; Liu, G. C.; Huang, Z.; Lv, S. X.; Shao, Y. L.; Lei, T. Continuous production of ultratough semiconducting polymer fibers with high electronic performance. *Sci. Adv.* **2024**, *10*, 1–11.
 - 71 Sun, T. T.; Liang, Y. F.; Ning, N. Y.; Wu, H. G.; Tian, M. Strain-insensitive stretchable conductive fiber based on helical core with double-network hydrogel. *Adv. Fiber Mater.* **2025**, *7*, 882–893.
 - 72 Wang, W. B.; Guo, W. J.; Zhu, J. H.; Li, H.; Liu, S. Q.; Erden, F.; Li, P. C.; Yang, M.; He, C. B. Highly conductive wet-spun PANI/CNTs fibers for wearable energy harvesting and thermal sensing. *J. Colloid Interface Sci.* **2026**, *704*, 139289.
 - 73 Zhang, C.; Liu, Y. L.; Li, H.; Liu, S. Q.; Li, P. C.; Zhang, H.; He, C. B. Enhanced thermoelectric properties of carbon nanotubes/polyaniline fibers through engineering doping level and orientation. *Compos. Sci. Technol.* **2024**, *253*, 110660.
 - 74 Xu, C.; Yang, S. W.; Li, P. C.; Wang, H.; Li, H.; Liu, Z. T. Wet-spun PEDOT: PSS/CNT composite fibers for wearable thermoelectric energy harvesting. *Compos. Commun.* **2022**, *32*, 101179.
 - 75 Liu, L.; Chen, J.; Liang, L. R.; Deng, L.; Chen, G. M. A PEDOT: PSS thermoelectric fiber generator. *Nano Energy* **2022**, *102*, 107678.
 - 76 Lee, H. J.; Anoop, G.; Lee, H. J.; Kim, C.; Park, J. W.; Choi, J.; Kim, H.; Kim, Y. J.; Lee, E.; Lee, S. G.; Kim, Y. M.; Lee, J. H.; Jo, J. Y. Enhanced thermoelectric performance of PEDOT:PSS/PANI-CSA polymer multilayer structures. *Energy Environ. Sci.* **2016**, *9*, 2806–2811.
 - 77 Hanus, R.; Agne, M. T.; Rettie, A. J. E.; Chen, Z. W.; Tan, G. J.; Chung, D. Y.; Kanatzidis, M. G.; Pei, Y. Z.; Voorhees, P. W.; Snyder, G. J. Lattice softening significantly reduces thermal conductivity and leads to high thermoelectric efficiency. *Adv. Mater.* **2019**, *31*, 1900108.
 - 78 Venkatasubramanian, R.; Siivola, E.; Colpitts, T.; O'Quinn, B. Thin-film thermoelectric devices with high room-temperature figures of merit. *Nature* **2001**, *413*, 597–602.
 - 79 Weinbach, Q.; Nielsen, C. B.; Biniek, L. Multi length scale porosity as a playground for organic thermoelectric applications. *J. Mater. Chem. C* **2021**, *9*, 10173–10192.
 - 80 Chen, M. Y.; Zhu, J.; Zhang, K.; Zhou, H. K.; Gao, Y. F.; Fan, J.; Chen, R. X.; Wang, H. L. Carbon nanofiber/polyaniline composite aerogel with excellent electromagnetic interference shielding, low thermal conductivity, and extremely low heat release. *Nano-Micro Lett.* **2024**, *17*, 80.
 - 81 Zuo, G. Z.; Liu, X. J.; Fahlman, M.; Kemerink, M. High seebeck coefficient in mixtures of conjugated polymers. *Adv. Funct. Mater.* **2017**, *28*, 173280.
 - 82 Villalva, D. R.; Derewjanko, D.; Zhang, Y. C.; Liu, Y.; Bates, A.; Sharma, A.; Han, J. H.; Gibert-Roca, M.; Arteaga, O. Z.; Jang, S.; Moro, S.; Costantini, G.; Gu, X. D.; Kemerink, M.; Baran, D. Intermolecular-force-driven anisotropy breaks the thermoelectric trade-off in n-type conjugated polymers. *Nat. Mater.* **2025**, *24*, 1236–1244.